长颈鹿物理学讲义-宇宙学



作者：长颈鹿 Gavin

July 24, 2026

Contents

1 引言 **3**

2 膨胀的宇宙 **4**
2.1 宇宙中的结构 4
2.2 熵密度 4
2.3 宇宙组分清单 4
2.4 距离 5

3 爱因斯坦场方程 **6**

4 玻尔兹曼方程 **7**

5 零阶爱因斯坦场方程 **8**
5.1 连续性方程 8
5.2 FLRW 度规 8
5.3 零阶爱因斯坦场方程（弗里德曼方程） 8

6 零阶玻尔兹曼方程 **10**
6.1 零阶玻尔兹曼方程 10
6.2 低温下的反应 10

7 均匀宇宙的推论 **11**
7.1 弗里德曼方程的推论 11
7.2 大爆炸核合成 11
7.3 复合时期 11

8 一阶爱因斯坦场方程 **13**
8.1 度规 13
8.2 能量-动量张量 13
8.3 一阶爱因斯坦场方程 14

9 一阶玻尔兹曼方程 **15**
9.1 光子 15
9.1.1 各向异性 15
9.1.2 极化 16
9.2 冷暗物质（CDM） 17
9.3 重子 17
9.4 中微子 17

10 初始条件 **18**
10.1 慢滚暴胀模型 18
10.2 初始扰动 18
10.3 初始功率谱 18

11 线性理论总结 **19**
11.1 线性方程组 19
11.2 初始条件 19

12 线性结果：物质与引力势 **21**

13 线性结果：光子（CMB） **22**
13.1 各向异性 22
13.2 极化 24

14 示踪天体 **27**
14.1 红移空间畸变与 AP 效应 27
14.2 角关联 28
14.3 热苏尼亚耶夫-泽尔多维奇效应 29

15 微扰理论 **31**

16 球坍缩 **33**
16.1 球坍缩方程 33
16.2 球坍缩的解 33
16.3 暗晕密度扰动 33
16.4 不同阶段的扰动 34

17 暗晕模型 **36**
17.1 暗晕密度轮廓 36
17.2 暗晕质量函数 37
17.3 暗晕的成团性 39
17.4 暗晕模型下的功率谱 40

18 星系团 **41**
18.1 质量-温度关系 41
18.2 顶帽窗函数下的功率谱 42

19 引力透镜 **43**
19.1 偏折角 43
19.2 质点引力透镜 44
19.3 星系形状 44
19.4 透镜/剪切角功率谱 45
19.5 剪切交叉角功率谱 46
19.6 CMB 透镜 47

20 数据分析 **49**
20.1 概述：从原始数据到参数约束 49
20.2 似然函数 49
20.3 示例：CMB 功率谱 50
20.4 示例：星系功率谱 52

A 数学基础 **55**
A.1 特殊函数 55
A.2 变换 55

1 引言

本文主要参考 Scott Dodelson《Modern Cosmology》(第二版) [2]
爱因斯坦场方程: "物质决定时空如何弯曲。"
玻尔兹曼方程: "时空决定物质如何运动。"

零阶爱因斯坦场方程: 弗里德曼方程。
零阶玻尔兹曼方程: 粒子演化方程。

一阶爱因斯坦场方程: 泊松方程 $\leftrightarrow$ 牛顿引力定律。
一阶玻尔兹曼方程: 1. 连续性方程 $\leftrightarrow$ 物质守恒; 2. 欧拉方程 $\leftrightarrow$ 牛顿第二定律

2 膨胀的宇宙

2.1 宇宙中的结构

物质功率谱:

$$\langle\delta(\mathbf{k_1})\delta(\mathbf{k_2})\rangle = (2\pi)^3\delta_D^{(3)}(\mathbf{k_1}+\mathbf{k_2})P(k_1) \tag{1}$$

关联函数:

$$\xi(\mathbf{x}-\mathbf{x}') = \langle\delta(\mathbf{x_1})\delta(\mathbf{x_2})\rangle = \int \frac{d^3k}{(2\pi)^3}P(k)e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} \tag{2}$$

2.2 熵密度

$$s = \frac{\rho+P}{T} \propto \frac{1}{a^3} \tag{3}$$

$$s = \frac{2\pi^2}{45}(\sum_{i=bosons} g_iT_i^3 + \frac{7}{8}\sum_{i=fermions} g_iT_i^3) \tag{4}$$

2.3 宇宙组分清单

光子: $\Omega_{\gamma 0}h^2 = 2.47\times10^{-5}$
重子: $\Omega_{b0}h^2 = 0.0225$
物质: $\Omega_{m0}h^2 = 0.1431$
中微子: $\Omega_{\nu 0}h^2 = \sum_i m_{\nu i}/94eV$

费米子与玻色子分布:

$$f_{FD}(E) = \frac{1}{e^{(E-\mu)/T}+1}, \quad f_{BE}(E) = \frac{1}{e^{(E-\mu)/T}-1} \tag{5}$$

密度:

$$\rho_s = \begin{cases} m_sn_s & ,\text{Matter} \\ g_s\frac{\pi^2}{30}T^4 & ,\text{Radiation Bosons} \\ \frac{7}{8}g_s\frac{\pi^2}{30}T^4 & ,\text{Radiation Fermions} \end{cases} \tag{6}$$

$$\Omega_{s0} = \frac{\rho_{s0}}{\rho_{cr0}} \tag{7}$$

温度:

$$T_s \propto \begin{cases} \frac{1}{a} & ,(m_s << T) \\ \frac{1}{a^2} & ,(m_s >> T) \end{cases} \tag{8}$$

2.4 距离

针对不同曲率 K 定义一个特殊函数：

$$\sin_k(\chi) = \begin{cases} K^{-1/2}\sin(\sqrt{K}\chi), & K > 0 \\ \chi, & K = 0 \\ (-K)^{-1/2}\sinh(\sqrt{-K}\chi), & K < 0 \end{cases} \tag{9}$$

共动距离：

$$\chi(t) = \int_t^{t_0} \frac{dt'}{a(t')} = \int_a^1 \frac{da_1}{a_1^2 H(a_1)} = \int_0^z \frac{dz_1}{H(z_1)} \tag{10}$$

角直径距离：

$$d_A = \frac{l}{\theta} = a\sin_k(\chi) = \frac{\sin_k(\chi)}{1+z} \tag{11}$$

光度距离：

$$d_L = \frac{\sin_k(\chi)}{a} = (1+z)\sin_k(\chi) \tag{12}$$

流量：

$$F = \frac{La^2}{4\pi\sin_k^2(\chi)(a)} = \frac{L}{4\pi d_L^2} \tag{13}$$

3 爱因斯坦场方程

爱因斯坦场方程:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G T_{\mu\nu} \tag{14}$$

联络:

$$\Gamma^{\mu}_{\alpha\beta} = \frac{1}{2} g^{\mu\nu} \left(g_{\nu\alpha,\beta} + g_{\nu\beta,\alpha} - g_{\alpha\beta,\nu} \right) \tag{15}$$

里奇张量:

$$R_{\mu\nu} = \Gamma^{\alpha}_{\mu\nu,\alpha} - \Gamma^{\alpha}_{\mu\alpha,\nu} + \Gamma^{\alpha}_{\beta\alpha}\Gamma^{\beta}_{\mu\nu} - \Gamma^{\alpha}_{\beta\nu}\Gamma^{\beta}_{\mu\alpha} \tag{16}$$

测地线方程:

$$\frac{d^2 x^{\mu}}{d\lambda^2} + \Gamma^{\mu}_{\alpha\beta} \frac{dx^{\alpha}}{d\lambda} \frac{dx^{\beta}}{d\lambda} = 0 \tag{17}$$

4 玻尔兹曼方程

玻尔兹曼方程：

$$C[f]=\frac{df}{dt}=\frac{\partial f}{\partial t}+\frac{\partial f}{\partial x^i}\frac{dx^i}{dt}+\frac{\partial f}{\partial p}\frac{dp}{dt}+\frac{\partial f}{\partial \hat{p}^i}\frac{d\hat{p}^i}{dt} \tag{18}$$

对于反应：$1+2\longleftrightarrow 3+4$

碰撞项：

$$\begin{aligned}C[f_1(\mathbf{p})]=&\frac{1}{2E_1(p)}\int\frac{d^3q}{(2\pi)^3 2E_2(q)}\int\frac{d^3p'}{(2\pi)^3 2E_3\left(p'\right)}\int\frac{d^3q'}{(2\pi)^3 2E_4\left(q'\right)}\\&\times|M|^2(2\pi)^4\delta_0^{(3)}\left(\mathbf{p}+\mathbf{q}-\mathbf{p}'-\mathbf{q}'\right)\delta_D^{(1)}\left(E_1+E_2-E_3-E_4\right)\\&\times\left\{f_3(\mathbf{p}')f_4(\mathbf{q}')\left[1\pm f_1(\mathbf{p})\right]\left[1\pm f_2(\mathbf{q})\right]-f_1(\mathbf{p})f_2(\mathbf{q})\left[1\pm f_3(\mathbf{p}')\right]\left[1\pm f_4(\mathbf{q}')\right]\right\}\end{aligned} \tag{19}$$

其中，$+$ 对应玻色子，$-$ 对应费米子

5 零阶爱因斯坦场方程

5.1 连续性方程

由：

$$\nabla_\mu T^\mu_\nu = \frac{\partial T^\mu_\nu}{\partial x^\mu} + \Gamma^\mu_{\alpha\mu} T^\alpha_\nu - \Gamma^\alpha_{\mu\nu} T^\mu_\alpha = 0 \tag{20}$$

可得：

$$\frac{\partial \rho}{\partial t} + 3H(\rho + P) = 0 \tag{21}$$

令：

$$\omega_s = \frac{P_s}{\rho_s} \tag{22}$$

得到：

$$\rho_s \propto a^{-3(1+\omega_s)} \tag{23}$$

不同组分的物态参数：

$$\omega_m = 0, \omega_r = \frac{1}{3}, \omega_K = -\frac{1}{3}, \omega_\Lambda = -1 \tag{24}$$

5.2 FLRW 度规

FLRW 度规：

$$ds^2 = -dt^2 + a^2(t)[d\chi^2 + \sin_k^2(\chi)(d\theta^2 + \sin^2\theta d\phi^2)] \tag{25}$$

能量-动量张量：

$$T^\mu_\nu = \begin{pmatrix} -\rho & & & \\ & P & & \\ & & P & \\ & & & P \end{pmatrix} \tag{26}$$

5.3 零阶爱因斯坦场方程（弗里德曼方程）

弗里德曼方程组：

$$\begin{aligned} (\frac{\dot{a}}{a})^2 + \frac{K}{a^2} &= \frac{8\pi G}{3}\rho \\ \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}(\rho + 3P) \end{aligned} \tag{27}$$

弗里德曼方程组也可表示为：

$$\begin{aligned} (\frac{\dot{a}}{a})^2 = H^2(a) &= H_0^2(\frac{\Omega_{r0}}{a^4} + \frac{\Omega_{m0}}{a^3} + \frac{\Omega_{K0}}{a^2} + \Omega_{\Lambda 0}) \\ \frac{\ddot{a}}{a} &= -\frac{H_0^2}{2}(\frac{2\Omega_{r0}}{a^4} + \frac{\Omega_{m0}}{a^3} - 2\Omega_{\Lambda 0}) \end{aligned} \tag{28}$$

临界密度:

$$\rho_{cr0} = \frac{3H_0^2}{8\pi G} \tag{29}$$

曲率密度占比:

$$K = -H_0^2 \Omega_{K0} \tag{30}$$

曲率半径:

$$r_K = |K|^{-1/2} \tag{31}$$

参数取值:

$$\begin{gathered} K = 0 \\ H_0 = 100h \cdot km \cdot s^{-1} \cdot Mpc^{-1} \\ a_{eq} = \frac{\Omega_{r0}}{\Omega_{m0}} = 2.9 \times 10^{-4} \end{gathered} \tag{32}$$

对于动力学暗能量:

$$\Omega_{\Lambda 0} \to \Omega_{DE} \exp[3 \int_a^1 (1 + \omega(a)) \frac{da}{a}] \tag{33}$$

6 零阶玻尔兹曼方程

6.1 零阶玻尔兹曼方程

$$C[f] = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{p}{E}\frac{\hat{p}^i}{a}\frac{\partial f}{\partial x^i} - Hp\frac{\partial f}{\partial p} \tag{34}$$

$$\frac{dn(t)}{dt} + 3Hn(t) = \int \frac{d^3p}{(2\pi)^3} C[f] \tag{35}$$

$$a^{-3}\frac{d(na^3)}{dt} = \int \frac{d^3p}{(2\pi)^3} C[f] \tag{36}$$

6.2 低温下的反应

当 $T \ll E - \mu$ 时，反应 $1 + 2 \longleftrightarrow 3 + 4$ 满足：

$$a^{-3}\frac{d(n_1 a^3)}{dt} = n_1^{(0)} n_2^{(0)} \langle \sigma v \rangle \{ \frac{n_3 n_4}{n_3^{(0)} n_4^{(0)}} - \frac{n_1 n_2}{n_1^{(0)} n_2^{(0)}} \} \tag{37}$$

数密度：

$$n_s = n_s^{(0)} e^{\mu_s / T_s} = g_s \int \frac{d^3p}{(2\pi)^3} f(E) \tag{38}$$

$\mu_s = 0$ 时的数密度：

$$n_s^{(0)} = \begin{cases} g_s (\frac{m_s T_s}{2\pi})^{\frac{3}{2}} e^{-m_s/T_s}, (m_s >> T) \\ g_s \frac{T_s^3}{\pi^2}, (m_s << T) \end{cases} \tag{39}$$

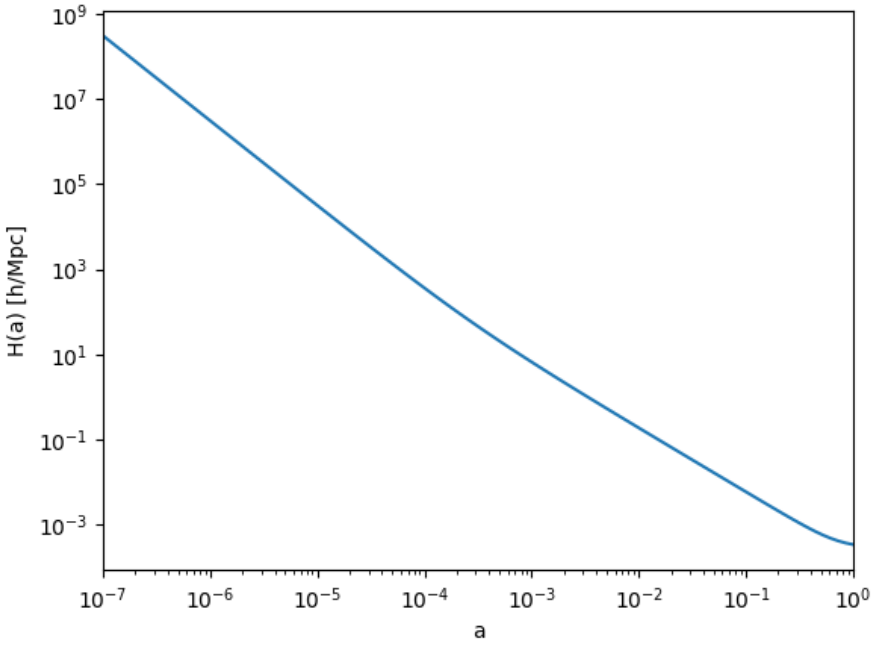


Figure 1: 哈勃参数 $H(a)$

7 均匀宇宙的推论

7.1 弗里德曼方程的推论

宇宙年龄:

$$t = \int_0^1 \frac{da}{aH(a)} = 13.8\, billion\ years \tag{40}$$

可观测宇宙半径:

$$r_{vi} = \int_0^1 \frac{da}{a^2 H(a)} = 46.3\, billion\ light\ years \tag{41}$$

哈勃参数的演化见图1
膨胀速率 $\dot{a} = H(z)/(1+z)$ 见图2与图3

7.2 大爆炸核合成

质子与中子的反应:

$$p + e^- \leftrightarrow n + \nu_e \tag{42}$$

结果见图4

7.3 复合时期

复合时期的反应:

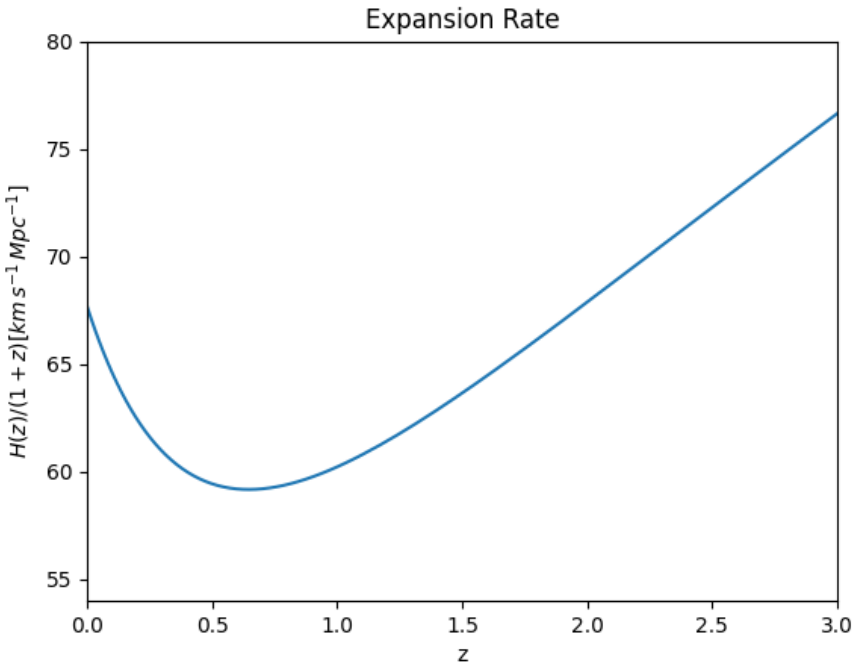


Figure 2: 膨胀速率

$$p + e^- \leftrightarrow H + \gamma \tag{43}$$

自由电子丰度:

$$X_e = \frac{n_e}{n_e + n_H} \tag{44}$$

X_e 的演化见图5

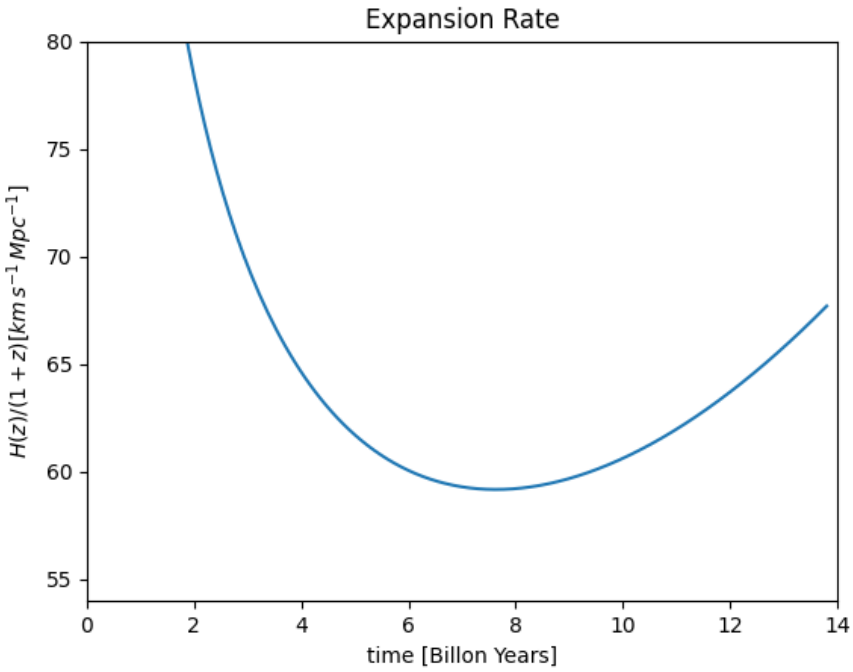


Figure 3: 膨胀速率

8 一阶爱因斯坦场方程

8.1 度规

标量扰动:

$$\begin{cases} g_{00} = -1 - 2\Psi \\ g_{0i} = 0 \\ g_{ij} = a^2(t)\delta_{ij}(1 + 2\Phi) \end{cases} \tag{45}$$

张量扰动:

$$\begin{cases} \delta g_{ij} = a^2(t) h_{ij}^{TT} \\ h_{ij}^{TT} = \begin{pmatrix} h_+ & h_\times & 0 \\ h_\times & -h_+ & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{cases} \tag{46}$$

8.2 能量-动量张量

$$\begin{aligned} T_\nu^\mu(\mathbf{x}, t) &= g \int \frac{d^3p}{(2\pi)^3} \frac{p^\mu p_\nu}{E(p)} f(\mathbf{x}, \mathbf{p}, t) \\ p^\mu &= [E(1 - \Psi), p^i(\frac{1 - \Phi}{a})] \\ -m^2 &= g_{\mu\nu} p^\mu p^\nu \end{aligned} \tag{47}$$

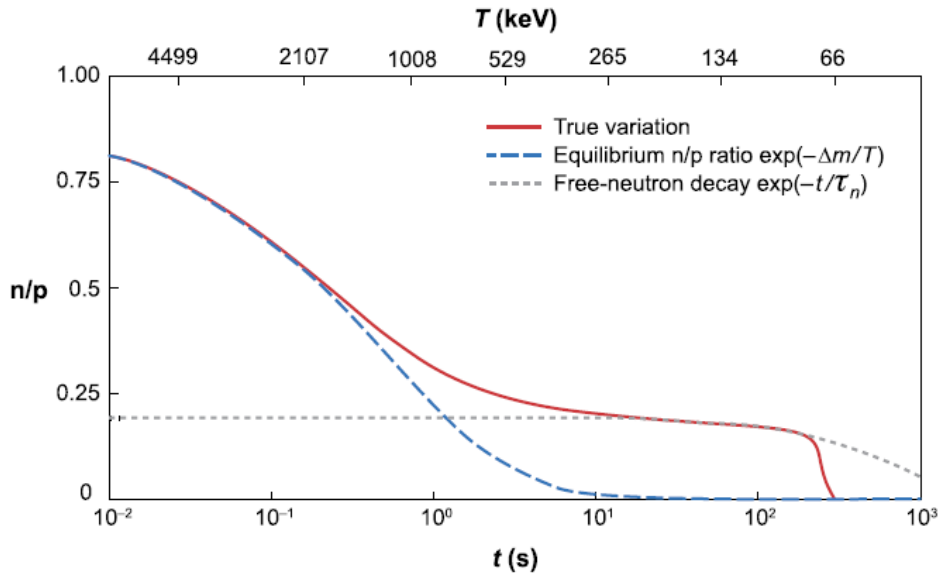


Figure 4: 大爆炸核合成（BBN）

$$
\begin{aligned}
T^0{}_0(x,t) &= -g\int \frac{d^3p}{(2\pi)^3}E(p)f(\mathbf{x},\mathbf{p},t)\\
T^0{}_i(x,t) &= ga(1+\Phi-\Psi)\int \frac{d^3p}{(2\pi)^3}p_i f(\mathbf{x},\mathbf{p},t)\\
T^i{}_j(x,t) &= g\int \frac{d^3p}{(2\pi)^3}\frac{p^i p_j}{E(p)}f(\mathbf{x},\mathbf{p},t)
\end{aligned}
\tag{48}
$$

8.3 一阶爱因斯坦场方程

标量扰动（泊松方程）：

$$
\begin{aligned}
k^2\Phi + 3\frac{a'}{a}\left(\Phi' - \Psi\frac{a'}{a}\right) &= 4\pi G a^2[\rho_{\mathrm{m}}\delta_{\mathrm{m}} + 4\rho_{\mathrm{r}}\Theta_{\mathrm{r},0}]\\
k^2(\Phi+\Psi) &= -32\pi G a^2\rho_{\mathrm{r}}\Theta_{\mathrm{r},2}\\
k^2\Phi = 4\pi\, G a^2&\left[\rho_{\mathrm{m}}\delta_{\mathrm{m}} + 4\rho_{\mathrm{r}}\Theta_{\mathrm{r},0} + \frac{3aH}{k}(i\rho_{\mathrm{m}}u_{\mathrm{m}} + 4\rho_{\mathrm{r}}\Theta_{\mathrm{r},1})\right].
\end{aligned}
\tag{49}
$$

张量扰动：

$$
h'' + 2\frac{a'}{a}h_t' + k^2 h_t = 0 \quad ,(t=+,\times)
\tag{50}
$$

物质（m）包含冷暗物质（c）与重子（b）。
辐射（r）包含光子（γ）与中微子（ν）。

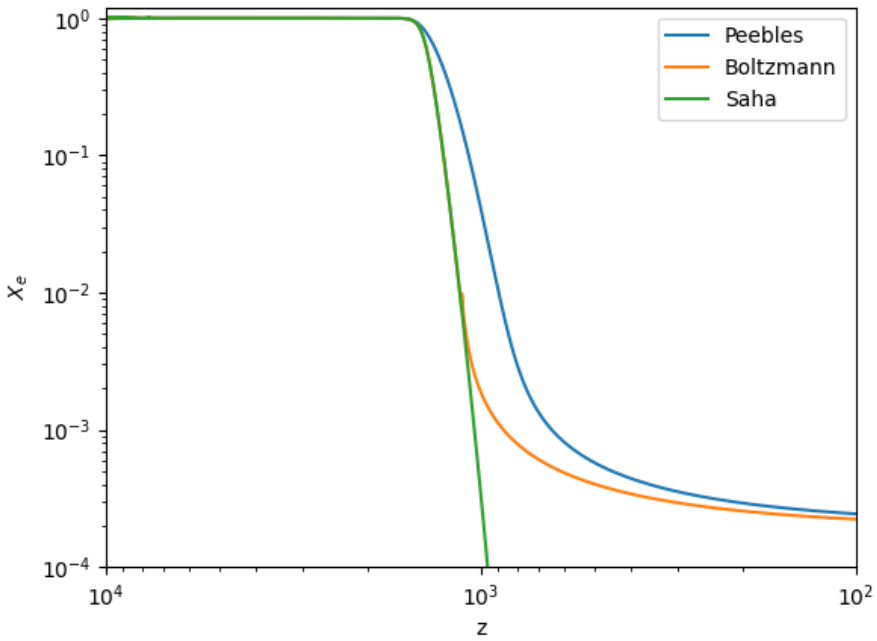


Figure 5: X_e

9 一阶玻尔兹曼方程

$$C[f] = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x^i}\frac{\hat{p}^i}{a}\frac{p}{E}(1 - \Phi + \Psi) - \frac{\partial f}{\partial p}\left\{\left[H + \dot{\Phi}\right]p + \frac{E}{a}\hat{p}^i\Psi_{,i}\right\} + \frac{\partial f}{\partial \hat{p}^i}\frac{E}{ap}\left[(\delta^{ik} - \hat{p}^i\hat{p}^k)(\frac{p^2}{E^2}\Phi - \Psi)_{,k}\right]. \tag{51}$$

9.1 光子

9.1.1 各向异性

温度扰动：

$$\Theta = \delta T/T \tag{52}$$

多极矩：

$$\Theta_l = \frac{1}{(-i)^l}\int_{-1}^{1}\frac{d\mu}{2}\mathcal{P}_l(\mu)\Theta(\mu, k, \eta) \tag{53}$$

光深：

$$\tau(\eta) = \int_{\eta}^{\eta_0} d\eta_1 n_e \sigma_T a \tag{54}$$

可见度函数：

$$g(\eta) = -\tau'(\eta)e^{-\tau(\eta)} \tag{55}$$

由可见度函数可得，复合发生在红移约 $z_{CMB} = 1080$ 处，见图6

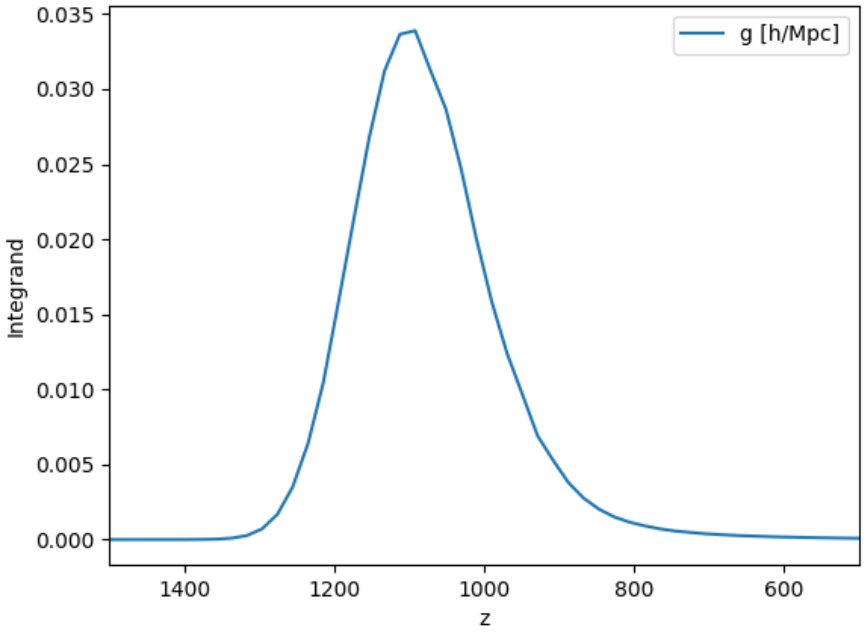


Figure 6: $g(z)$

光子各向异性的线性方程（CMB 各向异性）：

$$\begin{aligned} \Theta' + ik\mu\Theta &= -\Phi' - ik\mu\Psi - \tau'\left[\Theta_0 - \Theta + \mu u_{\rm b} - \frac{1}{2}\mathcal{P}_2(\mu)\Pi\right] \\ \Pi &= \Theta_2 + \Theta_{P2} + \Theta_{P0} \end{aligned} \tag{56}$$

各阶多极矩满足：

$$\begin{aligned} l = 0: &\quad \Theta_0' + k\Theta_1 = -\Phi', \\ l = 1: &\quad \Theta_1' - \frac{1}{3}k(\Theta_0 - 2\Theta_2) = \frac{k\Psi}{3} + \tau'\left[\Theta_1 - \frac{iu_{\rm b}}{3}\right]. \\ l = 2: &\quad \Theta_2' - \frac{1}{5}k(2\Theta_1 - 3\Theta_3) = (\frac{9}{10}\Theta_2 - \frac{\Theta_{P0}}{10} - \frac{\Theta_{P2}}{10})\tau' \\ l > 2: &\quad \Theta_l' + \frac{k}{(-i)^{l+1}}\int_{-1}^{1}\frac{d\mu}{2}\mu\mathcal{P}_l(\mu)\Theta(\mu) = \tau'\Theta_l \end{aligned} \tag{57}$$

9.1.2 极化

光子极化的线性方程（CMB 极化，E 模）：

$$\begin{aligned} \Theta_E' + ik\mu\Theta_E &= \tau'\left[\Theta_E - \frac{3}{4}(1-\mu^2)\Pi\right] \\ \Pi &= \Theta_2 + \Theta_{E2} + \Theta_{E0} \end{aligned} \tag{58}$$

9.2 冷暗物质（CDM）

冷暗物质扰动：

$$\delta_c = \delta n_c/\bar{n} \tag{59}$$

冷暗物质线性方程：

$$\begin{aligned} \delta_c' + iku_c &= -3\Phi' \quad \text{(Continuity Equation)} \\ u_c' + \frac{a'}{a}u_c &= -ik\Psi \quad \text{(Euler's Equation)} \end{aligned} \tag{60}$$

9.3 重子

重子扰动：

$$\delta_b = \delta\rho_e/\rho_e = \delta\rho_p/\rho_p \tag{61}$$

重子线性方程：

$$\begin{aligned} \delta_b' + iku_b &= -3\Phi' \quad \text{(Continuity Equation)} \\ u_b' + \frac{a'}{a}u_b &= -ik\Psi + \frac{\tau'}{R_b}(u_b + 3i\Theta_1) \quad \text{(Euler's Equation)} \end{aligned} \tag{62}$$

重子与光子的密度比：

$$R_b(\eta) = \frac{3\rho_b(\eta)}{4\rho_\gamma(\eta)} \tag{63}$$

9.4 中微子

中微子扰动：

$$\mathcal{N} = \delta T_\nu/T_\nu \tag{64}$$

中微子线性方程：

$$\mathcal{N}' + ik\mu\,\frac{p}{E_\nu(p)}\mathcal{N} - p\frac{a'}{a}\frac{\partial}{\partial p}\mathcal{N} = -\Phi' - ik\mu\frac{E_\nu(p)}{p}\Psi. \tag{65}$$

10 初始条件

10.1 慢滚暴胀模型

暴胀标量场：

$$\begin{aligned}\rho &= \frac{1}{2}\dot{\phi}^2 + V(\phi) \\ \mathcal{P} &= \frac{1}{2}\dot{\phi}^2 - V(\phi)\end{aligned} \tag{66}$$

慢滚参数：

$$\begin{aligned}\epsilon_{\rm sr} &\equiv \frac{d}{dt}\left(\frac{1}{H}\right) \\ \delta_{\rm sr} &\equiv \frac{1}{H}\frac{\ddot{\phi}}{\dot{\phi}}\end{aligned} \tag{67}$$

10.2 初始扰动

$$\begin{aligned}\Psi = -\frac{2}{3}\mathcal{R}, \Phi = \frac{2}{3}\mathcal{R}, \\ \Theta_0 = \frac{1}{3}\mathcal{R}, \mathcal{N}_0 = \frac{1}{3}\mathcal{R} \\ \delta_b = \mathcal{R}, \delta_c = \mathcal{R} \\ \Theta_1 = -\frac{k}{9aH}\mathcal{R}, \mathcal{N}_1 = -\frac{k}{9aH}\mathcal{R} \\ iu_b = -\frac{k}{3aH}\mathcal{R}, iu_c = -\frac{k}{3aH}\mathcal{R}\end{aligned} \tag{68}$$

10.3 初始功率谱

初始功率谱：

$$\begin{aligned}P_{\mathcal{R}}(k) &= \frac{2\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2\epsilon_{\rm Sr}}\bigg|_{aH=k} \equiv 2\pi^2\mathcal{A}_s k^{-3}\left(\frac{k}{k_{\rm P}}\right)^{n_s-1} \\ P_{\rm T}(k) &= 4P_h(k) = \frac{32\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2}\bigg|_{aH=k} \equiv 2\pi^2 A_{\rm T} k^{-3}\left(\frac{k}{k_{\rm p}}\right)^{n_{\rm T}}\end{aligned} \tag{69}$$

参数：

$$\begin{aligned}&n_s = 1 - 4\epsilon_{\rm sr} - 2\delta_{\rm sr} = 0.9682 \\ &n_{\rm T} = -2\epsilon_{\rm sr} \approx 0 \\ &k_p = 0.05 Mpc^{-1} \\ &ln(10^{10}A_s) = 3.0480\end{aligned} \tag{70}$$

初始张量与标量功率谱之比：

$$r(k) \equiv \frac{P_{\rm T}(k)}{P_{\mathcal{R}}(k)} \overset{k=k_{\rm p}}{=} \frac{\mathcal{A}_{\rm T}}{\mathcal{A}_s} = 16\epsilon_{\rm sr}\bigg|_{aH=k} \tag{71}$$

11 线性理论总结

11.1 线性方程组

标量扰动线性方程：

$$\begin{aligned} k^2\Phi + 3\frac{a'}{a}\left(\Phi' - \Psi\frac{a'}{a}\right) &= 4\pi G a^2[\rho_{\rm m}\delta_{\rm m} + 4\rho_{\rm r}\Theta_{\rm r,0}](\text{Poisson's Equation}) \\ k^2(\Phi+\Psi) &= -32\pi G a^2\rho_{\rm r}\Theta_{\rm r,2} \end{aligned} \tag{72}$$

张量扰动线性方程：

$$h'' + 2\frac{a'}{a}h_t' + k^2 h_t = 0 \quad ,(t=+,\times) \tag{73}$$

光子各向异性线性方程（CMB 各向异性）：

$$\begin{aligned} \Theta' + ik\mu\Theta &= -\Phi' - ik\mu\Psi - \tau'\left[\Theta_0 - \Theta + \mu u_{\rm b} - \frac{1}{2}\mathcal{P}_2(\mu)\Pi\right] \\ \Pi &= \Theta_2 + \Theta_{P2} + \Theta_{P0}. \end{aligned} \tag{74}$$

光子极化线性方程（CMB 极化，E 模）：

$$\begin{aligned} \Theta_E' + ik\mu\Theta_E &= \tau'\left[\Theta_E - \frac{3}{4}(1-\mu^2)\Pi\right] \\ \Pi &= \Theta_2 + \Theta_{E2} + \Theta_{E0}. \end{aligned} \tag{75}$$

冷暗物质线性方程：

$$\begin{aligned} \delta_c' + iku_c &= -3\Phi' \quad (\text{Continuity Equation}) \\ u_c' + \frac{a'}{a}u_c &= -ik\Psi \quad (\text{Euler's Equation}) \end{aligned} \tag{76}$$

重子线性方程：

$$\begin{aligned} \delta_b' + iku_b &= -3\Phi' \quad (\text{Continuity Equation}) \\ u_b' + \frac{a'}{a}u_b &= -ik\Psi + \frac{\tau'}{R_b}(u_b + 3i\Theta_1) \quad (\text{Euler's Equation}) \end{aligned} \tag{77}$$

中微子线性方程：

$$\mathcal{N}' + ik\mu\,\frac{p}{E_\nu(p)}\mathcal{N} - p\frac{a'}{a}\frac{\partial}{\partial p}\mathcal{N} = -\Phi' - ik\mu\frac{E_\nu(p)}{p}\Psi. \tag{78}$$

11.2 初始条件

初始条件：

$$
\begin{aligned}
\Psi = -\frac{2}{3}\mathcal{R}, \Phi = \frac{2}{3}\mathcal{R}, \\
\Theta_0 = \frac{1}{3}\mathcal{R}, \mathcal{N}_0 = \frac{1}{3}\mathcal{R} \\
\delta_b = \mathcal{R}, \delta_c = \mathcal{R} \\
\Theta_1 = -\frac{k}{9aH}\mathcal{R}, \mathcal{N}_1 = -\frac{k}{9aH}\mathcal{R} \\
iu_b = -\frac{k}{3aH}\mathcal{R}, iu_c = -\frac{k}{3aH}\mathcal{R}
\end{aligned}
\tag{79}
$$

初始功率谱:

$$
\begin{aligned}
P_{\mathcal{R}}(k) &= \frac{2\pi}{k^3}\frac{H^2}{m_{\mathrm{Pl}}^2\epsilon_{\mathrm{Sr}}}\bigg|_{aH=k} \equiv 2\pi^2\mathcal{A}_s k^{-3}\left(\frac{k}{k_{\mathrm{P}}}\right)^{n_s-1} \\
P_{\mathrm{T}}(k) &= 4P_h(k) = \frac{32\pi}{k^3}\frac{H^2}{m_{\mathrm{Pl}}^2}\bigg|_{aH=k} \equiv 2\pi^2 A_{\mathrm{T}} k^{-3}\left(\frac{k}{k_{\mathrm{p}}}\right)^{n_{\mathrm{T}}}
\end{aligned}
\tag{80}
$$

12 线性结果：物质与引力势

线性物质功率谱见图7、图8

$$P_L(k,a) = \left[\frac{2(k^2+3a^2H^2)}{5H_0^2\Omega_{m0}}T(k)D_+(a)\right]^2 P_{\mathcal{R}}(k) \tag{81}$$

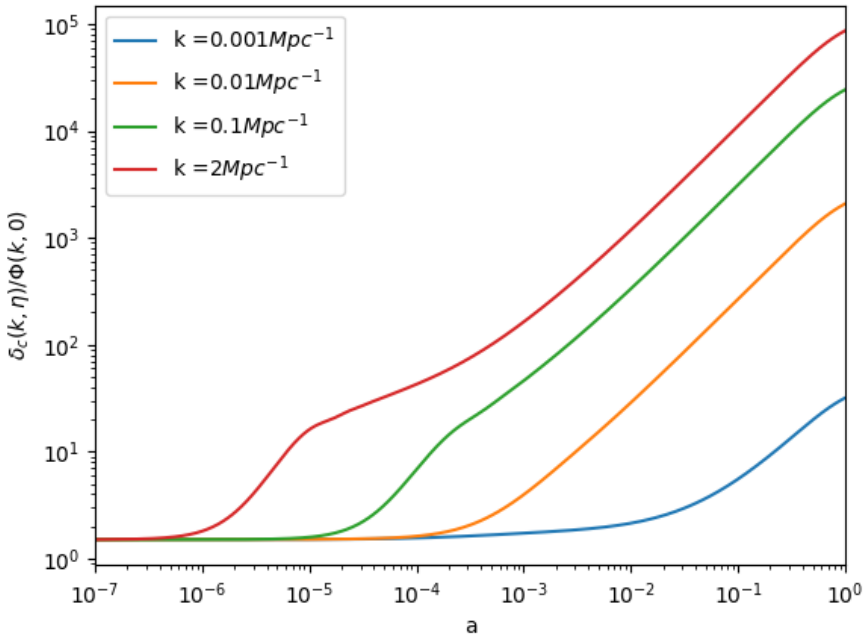


Figure 7: δ_c/Φ_0

线性引力势功率谱见图9

$$P_\Phi(k,a) = \left[\frac{3}{5}T(k)\frac{D_+(a)}{a}\right]^2 P_{\mathcal{R}}(k) \tag{82}$$

张量扰动见图10。小尺度模式衰减更早。
转移函数（BBKS 形式）：

$$T(k) = \frac{ln(1+2.34\cdot q)}{2.34\cdot q} \cdot \left[1+(3.89\cdot q)+(16.1\cdot q)^2+(5.46\cdot q)^3+(6.71\cdot q)^4\right]^{-1/4} \tag{83}$$

$$q(k) = \frac{k}{\Omega_{m0}h\cdot h/Mpc} \tag{84}$$

增长速率：

$$f(a) = \frac{dlnD_+(a)}{dlna} = \left[\frac{H_0^2}{H^2(a)}\frac{\Omega_{m0}}{a^3}\right]^{0.55} \tag{85}$$

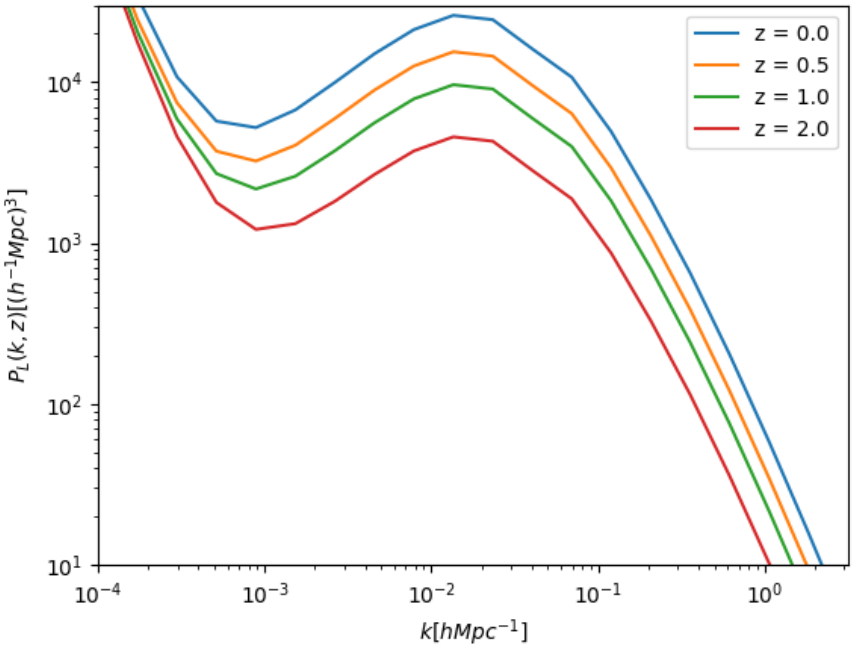


Figure 8: P_L

13 线性结果：光子（CMB）

13.1 各向异性

宇宙微波背景（CMB）各向异性功率谱见图11

$$C(l) = \frac{2}{\pi}\int_0^\infty dk k^2 P_{\mathcal{R}}(k)|T_l(k)|^2 \tag{86}$$

$$T_l(k) = \Theta_l(k,\eta_0)/\mathcal{R}(k) \tag{87}$$

重子与光子的密度比：

$$R_b(\eta) = \frac{3\rho_b(\eta)}{4\rho_\gamma(\eta)} \tag{88}$$

重子-光子流体的声速：

$$c_s(\eta) = \sqrt{\frac{1}{3[1+R_b(\eta)]}} \tag{89}$$

声视界：

$$r_s(\eta) = \int_0^\eta d\tilde{\eta} c_s(\tilde{\eta}) \tag{90}$$

光子的多极矩：

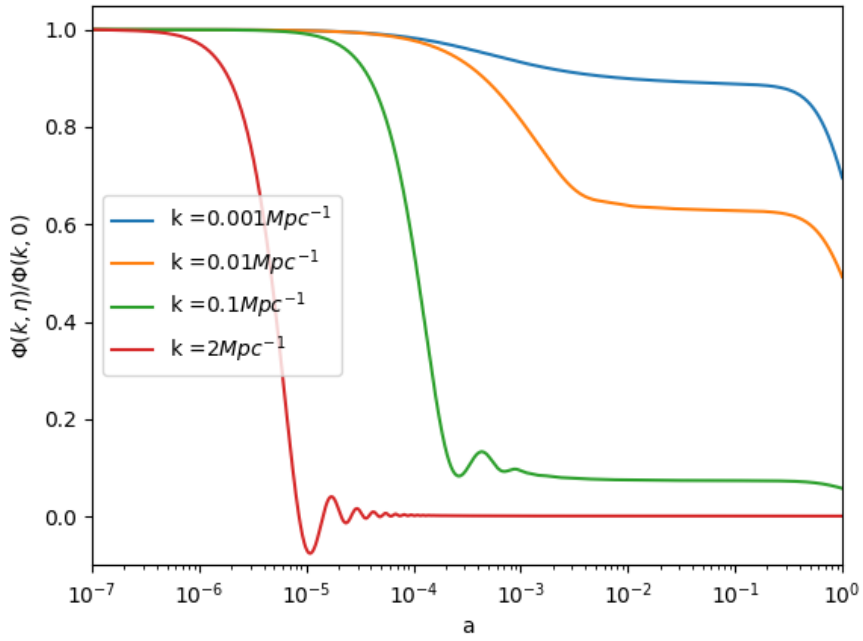


Figure 9: Φ/Φ_0

$$\begin{aligned}\Theta_l(k,\eta_0) \simeq & \left[\Theta_0(k,\eta_*)+\Psi(k,\eta_*)\right] j_l\left[k(\eta_0-\eta_*)\right] \\ & + 3\Theta_1(k,\eta_*)\left(j_{l-1}\left[k(\eta_0-\eta_*)\right]-(l+1)\frac{j_l\left[k(\eta_0-\eta_*)\right]}{k(\eta_0-\eta_*)}\right) \\ & + \int_0^{\eta_0} d\eta e^{-\tau}\left[\Psi'(k,\eta)-\Phi'(k,\eta)\right] j_l\left[k(\eta_0-\eta)\right]\end{aligned} \tag{91}$$

单极矩与偶极矩：

$$\begin{aligned}\Theta_0(\mathbf{k},\eta) = & \{\left[\Theta_0(\mathbf{k},0)+\Phi(\mathbf{k},0)\right]\cos(kr_s)-\Phi(\mathbf{k},\eta) \\ & + \frac{k}{\sqrt{3}}\int_0^{\eta} d\tilde{\eta}[\Phi(\mathbf{k},\tilde{\eta})-\Psi(\mathbf{k},\tilde{\eta})]\sin[k(r_s(\eta)-r_s(\tilde{\eta}))]\}\cdot\exp(-\frac{k^2}{k_D^2})\end{aligned} \tag{92}$$

$$\begin{aligned}\Theta_1(\mathbf{k},\eta) = & \{\frac{1}{\sqrt{3}}[\Theta_0(\mathbf{k},0)+\Phi(\mathbf{k},0)]\sin(kr_s) \\ & - \frac{k}{3}\int_0^{\eta} d\tilde{\eta}[\Phi(\mathbf{k},\tilde{\eta})-\Psi(\mathbf{k},\tilde{\eta})]\cos[k(r_s(\eta)-r_s(\tilde{\eta}))]\}\cdot\exp(-\frac{k^2}{k_D^2})\end{aligned} \tag{93}$$

Wayne Hu 近似下的单极矩与偶极矩（arxiv:9504057）：

$$\frac{\Theta_0(k,\eta)}{\mathcal{R}(k)} = \left[-\frac{1}{5}(1+3\mathcal{R}_b)\cos(kr_s(\eta))+(\mathcal{R}_b+1)\frac{\Phi(k,\eta)}{\mathcal{R}(k)}\right]\cdot\exp(-\frac{k^2}{k_D^2}) \tag{94}$$

$$\frac{\Theta_1(k,\eta)}{\mathcal{R}(k)} = \left[-\frac{1}{5}(1+3\mathcal{R}_b)c_s\cdot\sin(kr_s(\eta))\right]\cdot\exp(-\frac{k^2}{k_D^2}) \tag{95}$$

扩散阻尼：

$$\frac{1}{k_D^2(\eta)} \equiv \int_0^{\eta}\frac{d\tilde{\eta}}{6(1+R_b)n_e\sigma_{\mathrm{T}}a(\tilde{\eta})}\left[\frac{R_b^2}{1+R_b}+\frac{8}{9}\right] \tag{96}$$

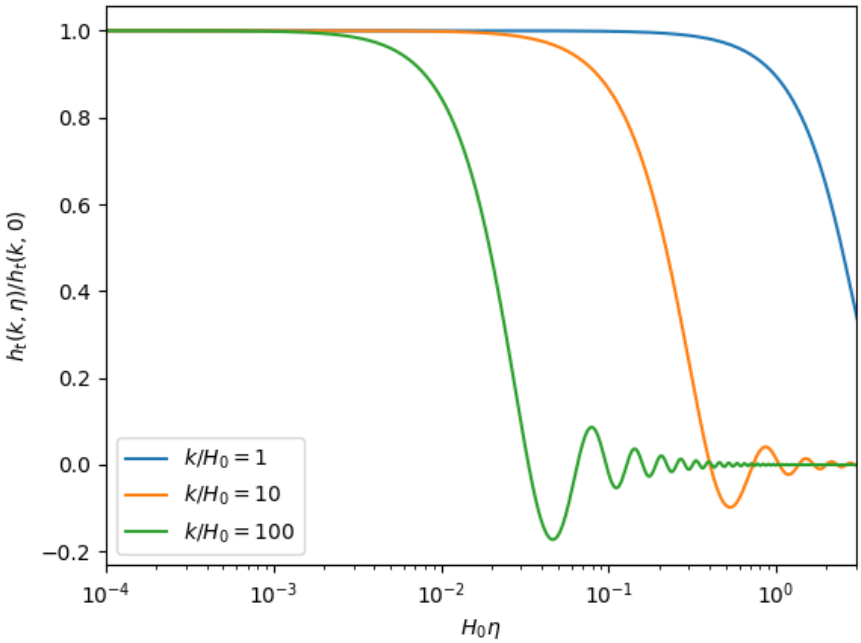


Figure 10: 张量扰动

13.2 极化

(1) E 模与 B 模
极化张量:

$$I_{ij} = \begin{pmatrix} I+Q & U \\ U & I-Q \end{pmatrix} \tag{97}$$

极化分量 Q、U 的含义见图12
将 Q、U 转换为旋转不变的极化分量 E、B，见图13

$$\begin{aligned} \nabla^2 E &= \partial_i \partial_j (I_{ij} - I\delta_{ij}) \\ \nabla^2 B &= \epsilon_{ik} \partial_j \partial_k (I_{ij} - I\delta_{ij}) \end{aligned} \tag{98}$$

其中，

$$\epsilon_{ij} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{99}$$

在傅里叶空间中:

$$\begin{aligned} E(\mathbf{l}) &= (\frac{l^i l^j}{l^2} - \frac{1}{2}\delta_{ij}) I_{ij} \\ B(\mathbf{l}) &= \epsilon_{ik} \frac{l^j l^k}{l^2} I_{ij} \end{aligned} \tag{100}$$

定义 $\mathbf{l}$ 为:

$$\mathbf{l} = l(\cos\phi_l, \sin\phi_l) \tag{101}$$

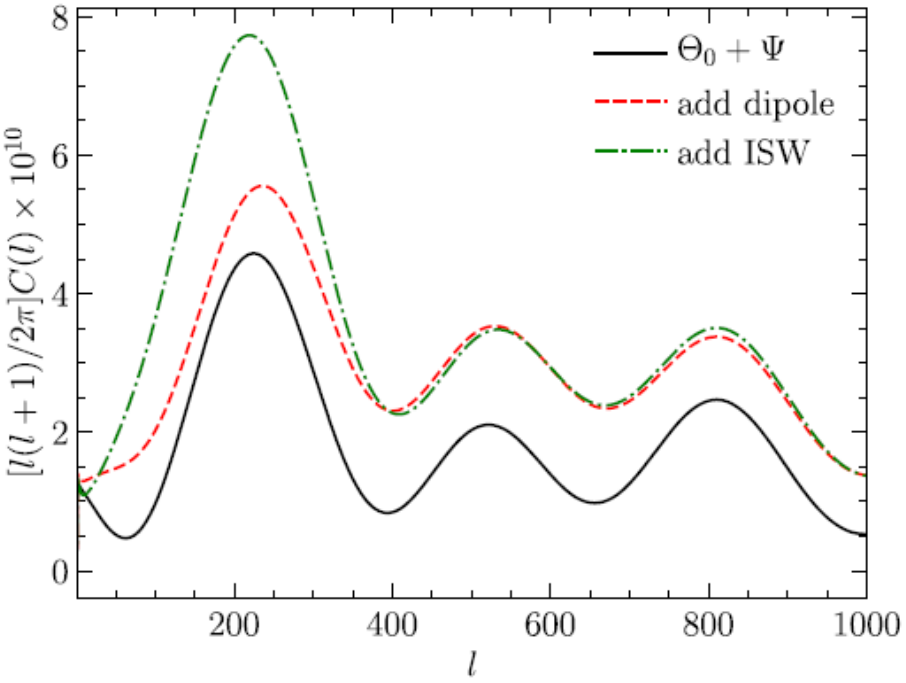


Figure 11: CMB 各向异性

可得 Q、U 与 E、B 的关系：

$$\begin{pmatrix} E(\mathbf{l}) \\ B(\mathbf{l}) \end{pmatrix} = \begin{pmatrix} \cos(2\phi_l) & \sin(2\phi_l) \\ -\sin(2\phi_l) & \cos(2\phi_l) \end{pmatrix} \begin{pmatrix} Q(\mathbf{l}) \\ U(\mathbf{l}) \end{pmatrix} \tag{102}$$

对矩阵求逆可得：

$$\begin{pmatrix} Q(\mathbf{l}) \\ U(\mathbf{l}) \end{pmatrix} = \begin{pmatrix} \cos(2\phi_l) & -\sin(2\phi_l) \\ \sin(2\phi_l) & \cos(2\phi_l) \end{pmatrix} \begin{pmatrix} E(\mathbf{l}) \\ B(\mathbf{l}) \end{pmatrix} \tag{103}$$

可以用 I、E、B 表示极化张量 I_{ij}：

$$I_{ij}(\mathbf{l}) = I(\mathbf{l})\delta_{ij} + \begin{pmatrix} \cos(2\phi_l) & \sin(2\phi_l) \\ \sin(2\phi_l) & -\cos(2\phi_l) \end{pmatrix} E(\mathbf{l}) + \begin{pmatrix} -\sin(2\phi_l) & \cos(2\phi_l) \\ \cos(2\phi_l) & \sin(2\phi_l) \end{pmatrix} B(\mathbf{l}) \tag{104}$$

(2) 功率谱
对于不同模式：

$$s = \{T, E, B\} \tag{105}$$

对应的功率谱见图14

$$C_{s_1 s_2}(l) = \frac{2}{\pi} \int_0^\infty dk k^2 P_{\mathcal{R}}(k) |T_l^{s_1}(k) T_l^{s_2 *}(k)| \tag{106}$$

$$T_l^s(k) = \Theta_{s,l}(k, \eta_0)/\mathcal{R}(k) \tag{107}$$

E 模：

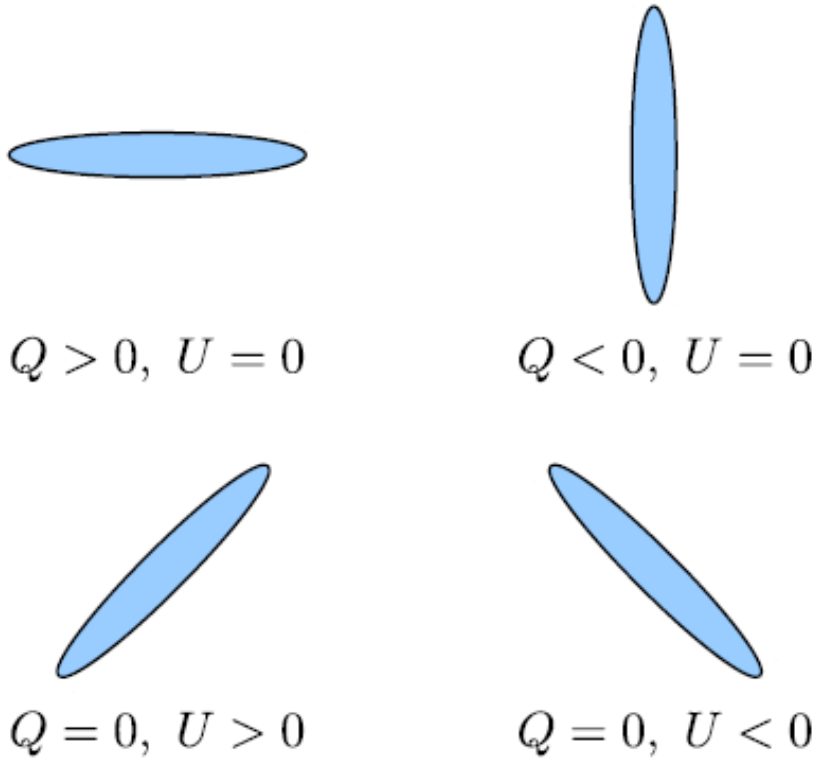


Figure 12: 极化分量：Q 与 U

$$\Theta_{E,l}(k,\eta_0) = -\frac{5k}{6\tau'}\Theta_1(k,\eta_*)\frac{l^2}{(k\eta_0)^2}j_l(k\eta_0) \tag{108}$$

B 模全部来自张量扰动：

$$B_\times^T(\mathbf{k}) \propto \frac{8\pi}{15}h_\times(\mathbf{k})(\hat{n}\cdot\hat{k}) \tag{109}$$

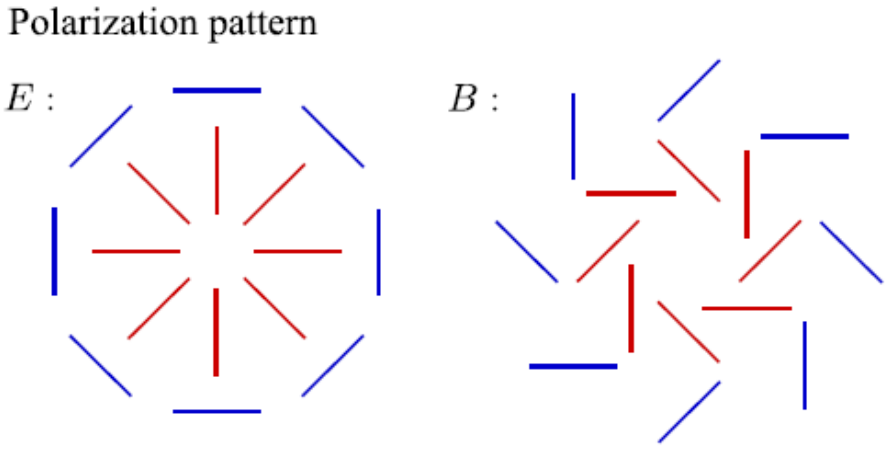


Figure 13: 旋转不变极化分量：E 与 B

14 示踪天体

14.1 红移空间畸变与 AP 效应

(1) 红移空间畸变（RSD）效应
RSD：由本动速度引起的功率谱畸变。

星系偏置：

$$\delta_g(\mathbf{x}, z) = b_1(z)\delta_m(\mathbf{x}, z) \tag{110}$$

视线方向与波矢的夹角余弦：

$$\mu_k = \widehat{\mathbf{e}}_z \cdot \widehat{\mathbf{k}} \tag{111}$$

(2) 阿尔科克-帕钦斯基（AP）效应
AP 效应：由宇宙学模型选取偏差引起的功率谱畸变。

基准宇宙学模型与真实值之间的共动距离和哈勃参数偏差：

$$\delta\chi = \chi_{fid} - \chi,\ \delta H = H_{fid} - H \tag{112}$$

真实波矢与观测波矢的关系：

$$\mathbf{k}[\mathbf{k}_{obs}] = ([1+\alpha_\perp]k^1_{\rm c,fid}, [1+\alpha_\perp]k^2_{\rm c,fid}, [1+\alpha_\parallel]k^3_{\rm c,fid}) \tag{113}$$

$$\alpha_\perp = \left.\frac{\delta\chi}{\chi_{fid}}\right|_{\bar{z}},\ \alpha_\parallel = -\left.\frac{\delta H}{H_{fid}}\right|_{\bar{z}} \tag{114}$$

(3) 观测星系功率谱（RSD+AP 效应）：

$$P_{\rm g,obs}(\mathbf{k}_{\rm obs}, \bar{z}) = P_{\rm L}(k, \bar{z})\left[b_1 + f\mu_k^2\right]^2 + P_N \tag{115}$$

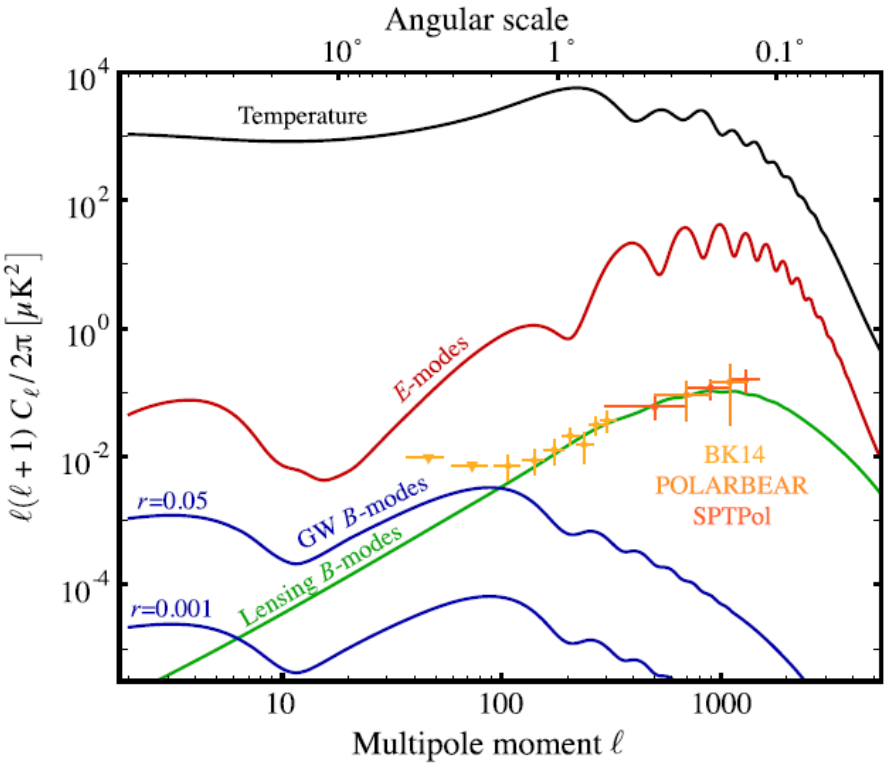


Figure 14: CMB 极化

更精确的结果如图15所示：

$$P_{\mathrm{g,obs}}(\mathbf{k}_{\mathrm{obs}},\bar{z}) = P_{\mathrm{L}}(k,\bar{z})\left[b_1 + f\mu_k^2\right]^2 D^{FOG}(k\mu_k) + P_N \tag{116}$$

$$D^{FOG}(k_z = k\mu_k) = \left|\langle \exp(i\frac{k_z v_z}{H})\rangle\right|^2 \tag{117}$$

14.2 角关联

星系距离分布：

$$W(\chi) = \frac{1}{N_g}\frac{dN_g}{d\chi} \tag{118}$$

投影过密度：

$$\Delta(\hat{\mathbf{n}}) = \int_0^\infty d\chi W(\chi)\delta_{g,obs}(\mathbf{x} = \hat{\mathbf{n}}\chi, \eta = \eta_0 - \chi) \tag{119}$$

角功率谱：

$$\langle \Delta_{g,lm}\Delta^*_{g,l'm'}\rangle = \delta_{ll'}\delta_{mm'}C_g(l) \tag{120}$$

角功率谱结果（Limber 近似）：

$$C_g(l) = \int \frac{d\chi}{\chi^2}W^2(\chi)P_{g,obs}\left(k = \frac{l+1/2}{\chi}, \mu_k = 0, \eta(\chi)\right) \tag{121}$$

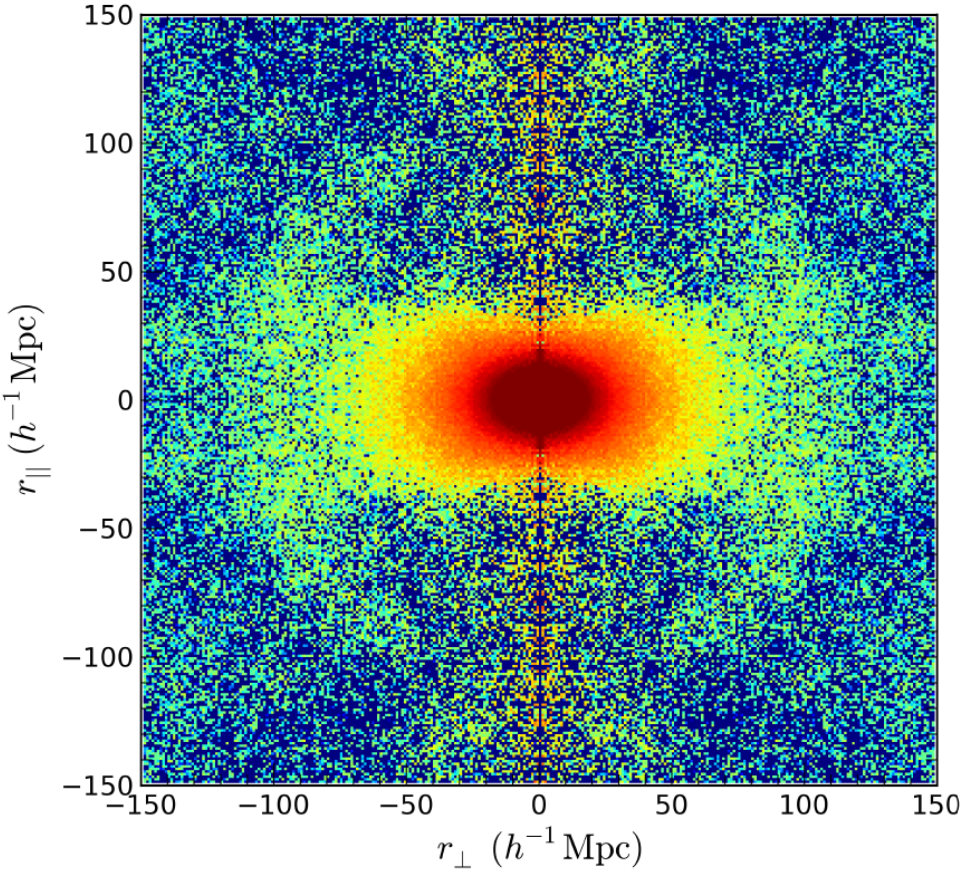


Figure 15: 红移空间畸变（RSD）

角关联函数：

$$w_g(\theta) = \int_0^\infty \frac{dl}{2\pi} l C_g(l) J_0(l\theta) \tag{122}$$

14.3 热苏尼亚耶夫-泽尔多维奇效应

热苏尼亚耶夫-泽尔多维奇（tSZ）效应：星系团高温气体中的电子对 CMB 光子的逆康普顿散射。

气体压强：

$$\mathcal{P}_{gas}(\mathbf{x},\eta) = n_e(\mathbf{x},\eta) T_e(\mathbf{x},\eta) = \overline{n_e T_e}(\eta)[1+\delta_{\mathcal{P}}(\mathbf{x},\eta)] \tag{123}$$

tSZ 的 y 参量是电离气体的动能密度/压强沿视线方向的积分：

$$y = \frac{\sigma_T}{m_e}\int n_e T_e a d\chi = \frac{\sigma_T}{m_e}\int \overline{n_e T_e}(\eta) a [1+\delta_{\mathcal{P}}(\mathbf{x},\eta)] d\chi \tag{124}$$

CMB 光子的分布：

$$\begin{cases} f(x,y) \overset{y\ll 1}{=} [1 + y\frac{1}{x^2}\frac{\partial}{\partial x}(x^4\frac{\partial}{\partial x})]\frac{1}{e^x-1} \\ x = p/T \end{cases} \tag{125}$$

$T_0 = 2.73\,\mathrm{K}$ 时的辐射强度见图16

$$
\begin{aligned}
u_\nu(T,x,y) &= 4\pi I_\nu(T,x,y) \\
I_\nu(T,x,y) &= 4\pi\nu^3 f(x,y) \\
\nu &= p/(2\pi) = x\cdot T/(2\pi)
\end{aligned}
\tag{126}
$$

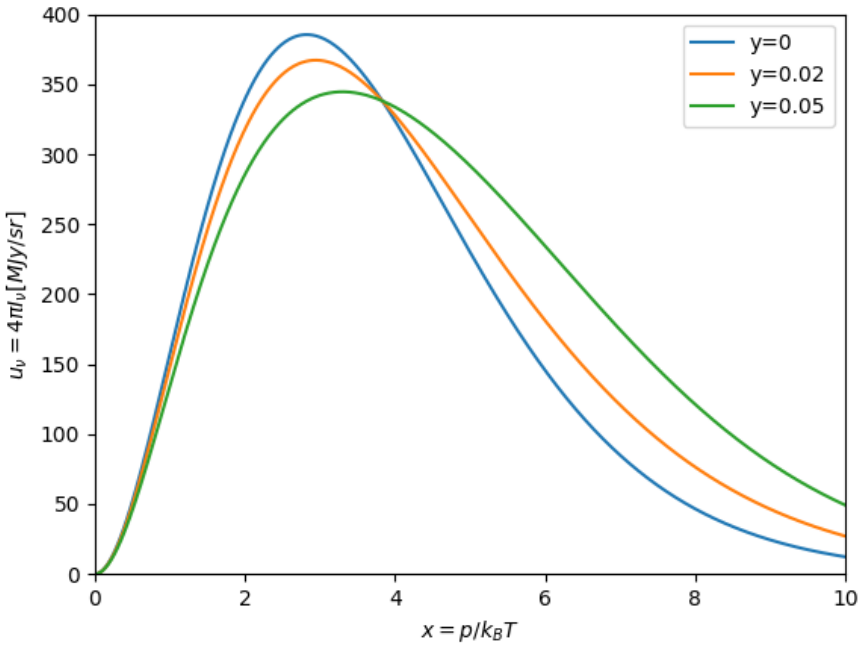


Figure 16: 热苏尼亚耶夫-泽尔多维奇效应

气体压强扰动的功率谱:

$$
\langle \delta_{\mathcal{P}}(\mathbf{k}_1)\delta_{\mathcal{P}}(\mathbf{k}_2)\rangle = (2\pi)^3\delta_D^{(3)}(\mathbf{k}_1+\mathbf{k}_2)P_{\mathcal{P}}(k_1) \tag{127}
$$

CMB 的 tSZ y 形变的角功率谱:

$$
C_y(l) = \left(\frac{\sigma_T}{m_e}\right)^2 \int_0^{\chi_*} \frac{d\chi}{\chi^2}[\overline{n_e T_e}a]^2 P_{\mathcal{P}}(k=\frac{l+1/2}{\chi},\eta(\chi)) \tag{128}
$$

15 微扰理论

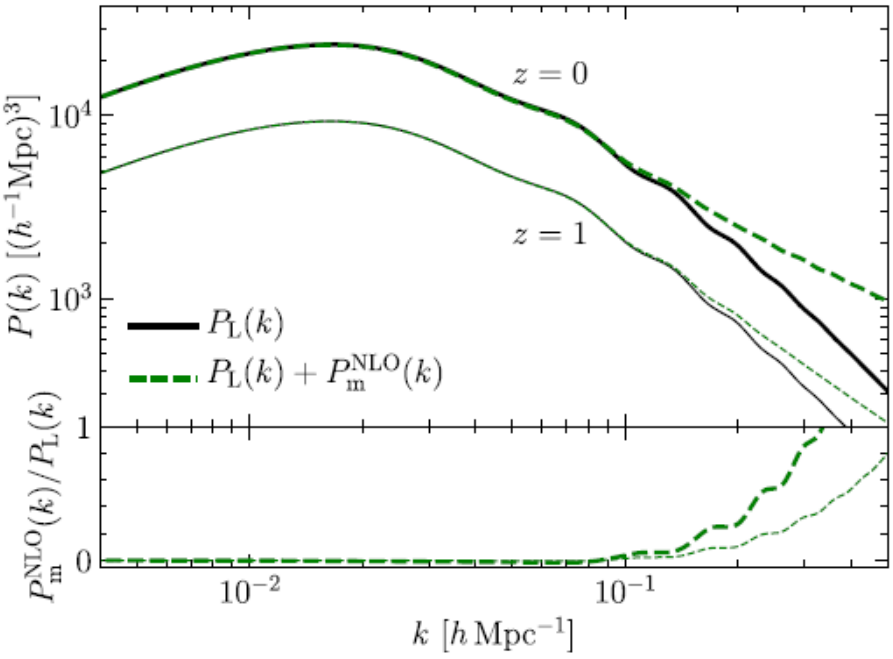


Figure 17: 微扰理论下的功率谱

弗拉索夫-泊松系统：

$$\begin{aligned}
&\frac{\partial f_{\rm m}}{\partial t} + \frac{\partial f_{\rm m}}{\partial x^j}\frac{p^j}{ma} - \frac{\partial f_{\rm m}}{\partial p^j}\left[Hp^j + \frac{m}{a}\frac{\partial \Psi}{\partial x^j}\right] = 0, \\
&\nabla^2\Psi = 4\pi G a^2\left[\int \frac{d^3p}{(2\pi)^3} f_{\rm m}(x,p,t) - \rho_{\rm m}(t)\right].
\end{aligned} \tag{129}$$

速度散度：

$$\theta_m = \partial_i u_m^i \tag{130}$$

由弗拉索夫-泊松系统可得微扰理论方程组：

$$\begin{aligned}
&{\delta_{\rm m}}' + \theta_{\rm m} = -\delta_{\rm m}\theta_{\rm m} - u_{\rm m}^j\frac{\partial}{\partial x^j}\delta_{\rm m}, \\
&{\theta_{\rm m}}' + aH\theta_{\rm m} + \nabla^2\Psi = -u_{\rm m}^j\frac{\partial}{\partial x^j}\theta_{\rm m} - (\partial_i u_{\rm m}^j)(\partial_j u_{\rm m}^i), \\
&\nabla^2\Psi = \frac{3}{2}\Omega_{\rm m}(\eta)(aH)^2\delta_{\rm m}.
\end{aligned} \tag{131}$$

物质密度与速度散度的扰动：

$$\begin{aligned}
\delta_{\rm m}(x,\eta) &= \delta^{(1)}(x,\eta) + \delta^{(2)}(x,\eta) + \cdots + \delta^{(n)}(x,\eta), \\
\theta_{\rm m}(x,\eta) &= \theta^{(1)}(x,\eta) + \theta^{(2)}(x,\eta) + \cdots + \theta^{(n)}(x,\eta),
\end{aligned} \tag{132}$$

各阶扰动的解：

$$
\begin{aligned}
\delta^{(n)}(k,\eta) &= D_+^n(\eta)\left[\prod_{i=1}^n\int\frac{d^3k_i}{(2\pi)^3}\right](2\pi)^3\delta_{\rm D}^{(3)}\left(\mathbf{k}-\sum_{i=1}^n\mathbf{k}_i\right)\\
&\quad\times F_n(\mathbf{k}_1,\cdots,\mathbf{k}_n)\delta_0(\mathbf{k}_1)\cdots\delta_0(\mathbf{k}_n),\\
\theta^{(n)}(k,\eta) &= -aHfD_+^n(\eta)\left[\prod_{i=1}^n\int\frac{d^3k_i}{(2\pi)^3}\right](2\pi)^3\delta_{\rm D}^{(3)}\left(\mathbf{k}-\sum_{i=1}^n\mathbf{k}_i\right)\\
&\quad\times G_n(\mathbf{k}_1,\cdots,\mathbf{k}_n)\delta_0(\mathbf{k}_1)\cdots\delta_0(\mathbf{k}_n).
\end{aligned}
\tag{133}
$$

其中:

$$
\begin{aligned}
F_1 &= G_1 = 1\\
F_2(\mathbf{k}_1,\mathbf{k}_2) &= \frac{5}{7}+\frac{2}{7}\frac{(\mathbf{k}_1\cdot\mathbf{k}_2)^2}{k_1^2k_2^2}+\frac{1}{2}\widehat{\mathbf{k}}_1\cdot\widehat{\mathbf{k}}_2\left(\frac{k_1}{k_2}+\frac{k_2}{k_1}\right)\\
G_2(\mathbf{k}_1,\mathbf{k}_2) &= \frac{3}{7}+\frac{4}{7}\frac{(\mathbf{k}_1\cdot\mathbf{k}_2)^2}{k_1^2k_2^2}+\frac{1}{2}\widehat{\mathbf{k}}_1\cdot\widehat{\mathbf{k}}_2\left(\frac{k_1}{k_2}+\frac{k_2}{k_1}\right)\\
&\cdots
\end{aligned}
\tag{134}
$$

功率谱见图17

$$
\begin{aligned}
&P(k,\eta) = P_{\rm L}(k,\eta)+P^{\rm NLO}(k,\eta)+\cdots,\\
&P^{\rm NLO}(k,\eta) = P^{(22)}(k,\eta)+2P^{(13)}(k,\eta),\\
&P^{(22)}(k,\eta) = 2\int\frac{d^3p}{(2\pi)^3}\Big[F_2(\mathbf{p},\mathbf{k}-\mathbf{p})\Big]^2P_{\rm L}(p,\eta)P_{\rm L}(|\mathbf{k}-\mathbf{p}|,\eta),\\
&P^{(13)}(k,\eta) = 3P_{\rm L}(k,\eta)\int\frac{d^3p}{(2\pi)^3}F_3(\mathbf{p},-\mathbf{p},\mathbf{k})P_{\rm L}(p,\eta).
\end{aligned}
\tag{135}
$$

双谱:

$$
\begin{aligned}
\langle\delta_{\rm m}(\mathbf{k}_1,\eta)\delta_{\rm m}(\mathbf{k}_2,\eta)\delta_{\rm m}(\mathbf{k}_3,\eta)\rangle &= (2\pi)^3\delta_{\rm D}^{(3)}(\mathbf{k}_1+\mathbf{k}_2+\mathbf{k}_3)\\
&\quad\times\left[2F_2(\mathbf{k}_1,\mathbf{k}_2)P_{\rm L}(k_1,\eta)P_{\rm L}(k_2,\eta)+2\text{perm.}\right]
\end{aligned}
\tag{136}
$$

16 球坍缩

16.1 球坍缩方程

爱因斯坦-德西特宇宙（$\Lambda = 0$，$k = 0$）的物质主导时期，忽略压强与重子效应。

牛顿引力方程：

$$\ddot{r} = -\frac{GM}{r} \tag{137}$$

能量守恒：

$$E = \frac{1}{2}\dot{r} - \frac{GM}{r} \tag{138}$$

扰动质量 M 与能量 E 为常数。

16.2 球坍缩的解

当 $E < 0$ 时，解为：

$$\begin{aligned} r(\theta) &= \frac{GM}{2|E|}(1 - \cos\theta) \\ t(\theta) &= \frac{GM}{|2E|^{3/2}}(\theta - \sin\theta) \end{aligned} \tag{139}$$

16.3 暗晕密度扰动

暗晕密度：

$$\rho_p(\theta) = \frac{M}{(4\pi/3)r^3} = \frac{6|E|^3}{\pi G^3 M^2}\frac{1}{(1-\cos\theta)^3} \tag{140}$$

由弗里德曼方程给出的背景密度：

$$\bar{\rho}(\theta) = \frac{3H^2}{8\pi G} = \frac{1}{6\pi G t^2} = \frac{4|E|^3}{3\pi G^3 M^2}\frac{1}{(\theta - \sin\theta)^2} \tag{141}$$

暗晕密度扰动：

$$1 + \delta_p(\theta) = \frac{\rho_p}{\bar{\rho}} = \frac{9}{2}\frac{(\theta - \sin\theta)^2}{(1-\cos\theta)^3} \propto \frac{t^2}{r^3} \tag{142}$$

线性密度扰动（$\theta \ll 1$）：

$$\delta_L(\theta) = \frac{3|E|}{10}\left(\frac{6}{GM}\right)^{2/3} t^{2/3} = \frac{3}{20}[6(\theta - \sin\theta)]^{2/3} \propto t^{2/3} \tag{143}$$

16.4 不同阶段的扰动

(1) 转向时刻：
时刻与半径：

$$\begin{aligned} \theta_{ta} &= \pi \\ t_{ta} = t(\theta_{ta}) &= \pi \cdot \frac{GM}{|2E|^{3/2}} \\ r_{ta} = r(\theta_{ta}) &= 2 \cdot \frac{GM}{2|E|} \end{aligned} \tag{144}$$

扰动：

$$\begin{aligned} \delta_{p,ta} &= \delta_p(\theta_{ta}) = 4.55 \\ \delta_{L,ta} &= \delta_L(\theta_{ta}) = 1.06 \end{aligned} \tag{145}$$

坍缩晚期 $r(t)$ 的关系不再成立，因为该时期重子压强不可忽略。
坍缩晚期，可通过转向时刻的扰动结合以下关系估算：

$$1 + \delta_p \propto \frac{t^2}{r^3}, \quad \delta_L \propto t^{2/3} \tag{146}$$

(2) 束缚时刻：
暗晕达到束缚状态的时刻。
时刻与半径：

$$\begin{aligned} \theta_c &= \frac{3}{2}\pi \\ t_c = t(\theta_c) &\approx 2t_{ta} \\ r_c = r(\theta_c) &= \frac{1}{2} r_{ta} \end{aligned} \tag{147}$$

扰动：

$$\begin{aligned} 1 + \delta_{p,c} &= \left(\frac{t_c}{t_{ta}}\right)^2 \left(\frac{r_{ta}}{r_c}\right)^3 (1 + \delta_{p,ta}) = 177.6 \\ \delta_{cr} = \delta_{L,c} &= \left(\frac{t_c}{t_{ta}}\right)^{2/3} \delta_{L,ta} = 1.686 \end{aligned} \tag{148}$$

(3) 位力化时刻：
暗晕达到位力平衡的时刻。
时刻与半径：

$$\begin{aligned} \theta_{vir} &= 3\pi \\ t_{vir} = t(\theta_{vir}) &= 3t_{ta} \\ r_{vir} &= \frac{1}{2} r_{ta} \end{aligned} \tag{149}$$

扰动：

$$
\begin{aligned}
1 + \delta_{p,vir} &= \left(\frac{t_{vir}}{t_{ta}}\right)^2 \left(\frac{r_{ta}}{r_{vir}}\right)^3 (1 + \delta_{p,ta}) = 400 \\
\delta_{L,vir} &= \left(\frac{t_{vir}}{t_{ta}}\right)^{2/3} \delta_{L,ta} = 2.20
\end{aligned}
\tag{150}
$$

17 暗晕模型

17.1 暗晕密度轮廓

暗晕的 Navarro-Frenk-White 密度轮廓（NFW 轮廓）见图18

$$\rho_{\rm h}(r) = \frac{\rho_s}{(r/r_{sh})(1+r/r_{sh})^2} \tag{151}$$

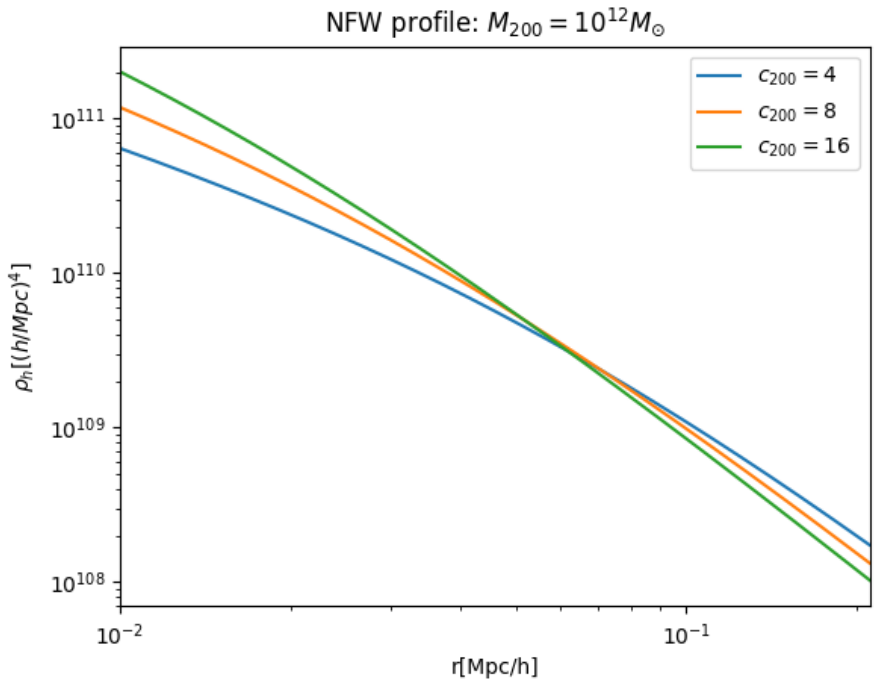


Figure 18: NFW 暗晕密度轮廓 $\rho_h(r)$

集中度:

$$c_\Delta = R_\Delta / r_{sh} \tag{152}$$

暗晕质量:

$$M_\Delta = M(< R_\Delta) = \frac{4\pi}{3} R_\Delta^3 \cdot \rho_{m0} \cdot \Delta \tag{153}$$

$$M_\Delta = \int_0^{R_\Delta} 4\pi r^2 \rho_h(r) dr \tag{154}$$

通常取 $\Delta = 200$。
对于给定的 Δ，可以用 M_Δ 和 c_Δ 表示 $\rho_h(r)$:

$$
\begin{aligned}
\rho_{\mathrm{h}}(r) &= \frac{\rho_s}{(r/r_{sh})(1+r/r_{sh})^2} \\
R_\Delta &= \left(\frac{3M_\Delta}{4\pi\rho_{m0}\cdot\Delta}\right)^{1/3} \\
r_{sh} &= R_\Delta/c_\Delta \\
\rho_s &= \frac{M_\Delta}{4\pi r_{sh}^3}\left[\ln(c_\Delta+1)+\frac{1}{c_\Delta+1}-1\right]^{-1}
\end{aligned}
\tag{155}
$$

k 空间下的 NFW 暗晕密度轮廓：

$$
\begin{aligned}
\rho_h\left(k=\frac{x}{r_s},M\right) = \frac{M}{\mathcal{N}}[&\cos x\,[\mathrm{Ci}([c+1]x)-\mathrm{Ci}(x)] \\
&+\sin x\,[\mathrm{Si}([c+1]x)-\mathrm{Si}(x)]-\frac{\sin cx}{(c+1)x}]
\end{aligned}
\tag{156}
$$

$$
\begin{aligned}
\mathcal{N} &= \frac{1}{c+1}+\log(c+1)-1 \\
r_{sh} &= R_{200}/c, \quad c=8
\end{aligned}
\tag{157}
$$

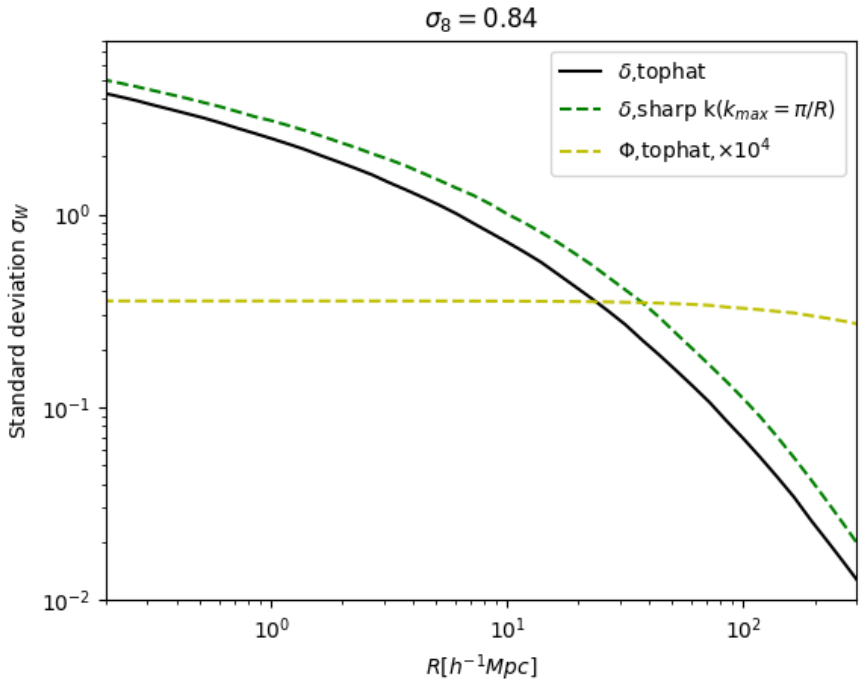


Figure 19: $\sigma_W(R)$

17.2 暗晕质量函数

拉格朗日半径（以平均物质密度包围暗晕质量的半径）：

$$
R_L(M) = 1.40h^{-1}\,\mathrm{Mpc}\left(\frac{M}{10^{12}h^{-1}M_\odot}\right)^{1/3} \tag{158}
$$

顶帽窗函数描述均匀密度的球形扰动：

$$W_R(x) = \begin{cases} \frac{3}{4\pi R^3}, & 0 \le x < R \\ 0, & x \ge R \end{cases} \tag{159}$$

通过傅里叶变换，可得傅里叶空间的顶帽窗函数：

$$W_R(k) = \frac{3}{(kR)^3}[\sin(kR) - (kR)\cos(kR)] \tag{160}$$

关联函数见图19

$$\begin{aligned} \sigma(M,z) &= \frac{D_+(z)}{D_+(z=0)}\sigma_W[R_L(M)] \\ \sigma_W^2(R) &= \frac{1}{2\pi^2}\int_0^\infty dk k^2 P_L(k)|W_R(k)|^2 \end{aligned} \tag{161}$$

以及 σ_8：

$$\sigma_8 = \sigma_W(R = 8Mpc/h) \tag{162}$$

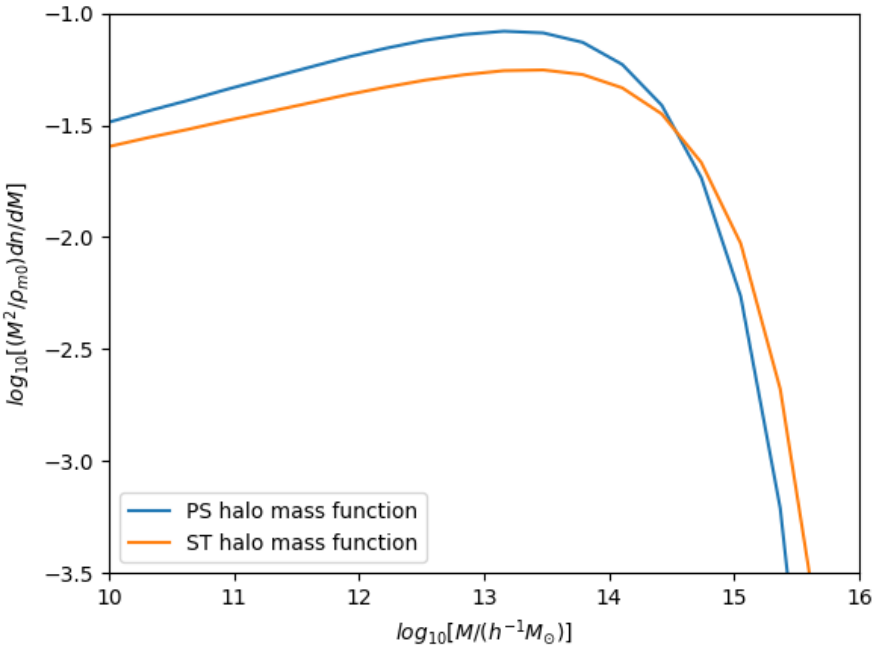


Figure 20: 暗晕质量函数

Press-Schechter（PS）暗晕质量函数：

$$\begin{aligned} \frac{dn(M,z)}{d\ln M} &= \frac{\rho_{\rm m0}}{M} f_{\rm PS}\left(\nu = \frac{\delta_{\rm cr}}{\sigma(M,z)}\right)\left|\frac{d\ln\sigma(M,z)}{d\ln M}\right| \\ f_{\rm PS}(\nu) &= \sqrt{\frac{2}{\pi}}\nu e^{-\nu^2/2}, \quad \delta_{cr} = 1.686 \end{aligned} \tag{163}$$

Sheth-Tormen（ST）暗晕质量函数：

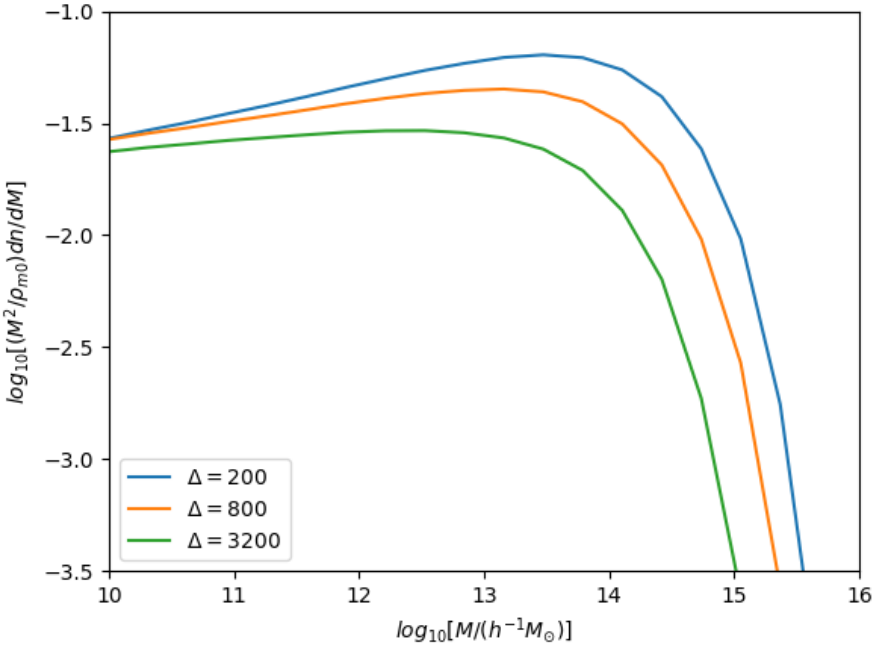


Figure 21: Tinker2008 暗晕质量函数

$$\frac{dn(M,z)}{d\ln M} = 0.322(1+\frac{1}{\nu'^{0.6}})\frac{\rho_{\rm m0}}{M} f_{\rm PS}(\nu')\left|\frac{d\ln\sigma(M,z)}{d\ln M}\right|$$
$$f_{\rm PS}(\nu') = \sqrt{\frac{2}{\pi}}\nu' e^{-\nu'^2/2}, \quad \delta_{cr} = 1.686 \tag{164}$$
$$\nu' = \sqrt{0.707}\nu = \sqrt{0.707}\frac{\delta_{cr}}{\sigma(M,z)}$$

PS 与 ST 暗晕质量函数见图20，Tinker2008 数值暗晕质量函数见图21。
质量大于 M 的暗晕数密度见图??

$$N(>M_1,z) = \int_{M_1}^{\infty} dn(M,z) = \int_{M_1}^{\infty}\frac{dn(M,z)}{d\ln M}d\ln M \tag{165}$$

17.3 暗晕的成团性

对于 PS 暗晕质量函数，在 $k = 0.01h/Mpc$ 处归一化满足 $P_{2h}(k) = P_L(k)$ 的暗晕-暗物质偏置：

$$\delta^{(1)}_{h,l}(\mathbf{x},t) = b_1(M,z)\delta^{(1)}_l(\mathbf{x},t)$$
$$b_1^{PS}(M,z) = 8.81\cdot\frac{\nu^2-1}{\delta_{cr}} \tag{166}$$
$$\nu = \delta_{cr}/\sigma(M,z)$$

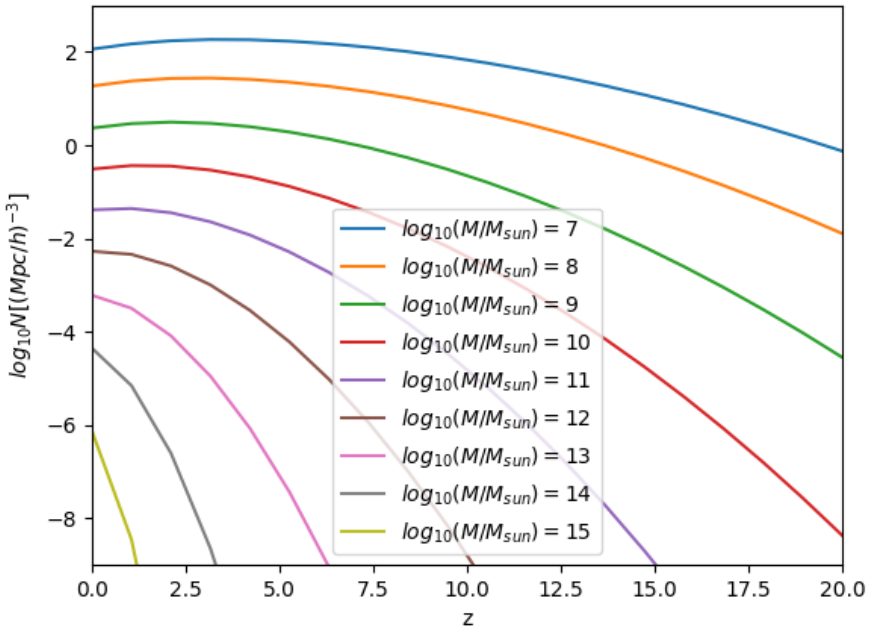


Figure 22: $N(>M,z)$

17.4 暗晕模型下的功率谱

在观测中，我们通过暗晕密度轮廓 $\rho_h(\mathbf{x}',M)$、暗晕质量函数 $dn/d\ln M$ 以及暗晕成团性 $\delta_h(\mathbf{x}',M)$ 得到暗物质扰动 $\delta_{\rm m}^{\rm HM}$。

暗晕模型中的暗物质扰动：

$$\delta_{\rm m}^{\rm HM}(x)=\int d\ln M\frac{1}{\rho_{\rm m0}}\frac{dn}{d\ln M}\int d^3x'\delta_h(\mathbf{x}',M)\rho_h(|\mathbf{x}-\mathbf{x}'|,M) \tag{167}$$

暗晕模型的功率谱（NFW 密度轮廓 +PS 暗晕质量函数 + 线性暗晕偏置）见图23

$$\begin{aligned}P^{\rm HM}(k,a)&=P_{\rm 2h}(k,a)+P_{\rm 1h}(k,a),\\ P_{\rm 2h}(k,a)&=\left[\frac{1}{\rho_{\rm m0}}\int d\ln M\frac{dn}{d\ln M}b_1(M,a)\rho_h(k,M)\right]^2P_{\rm L}(k,a),\\ P_{\rm 1h}(k,a)&=\frac{1}{\rho_{\rm m0}^2}\int d\ln M\frac{dn}{d\ln M}[\rho_h(k,M)]^2.\end{aligned} \tag{168}$$

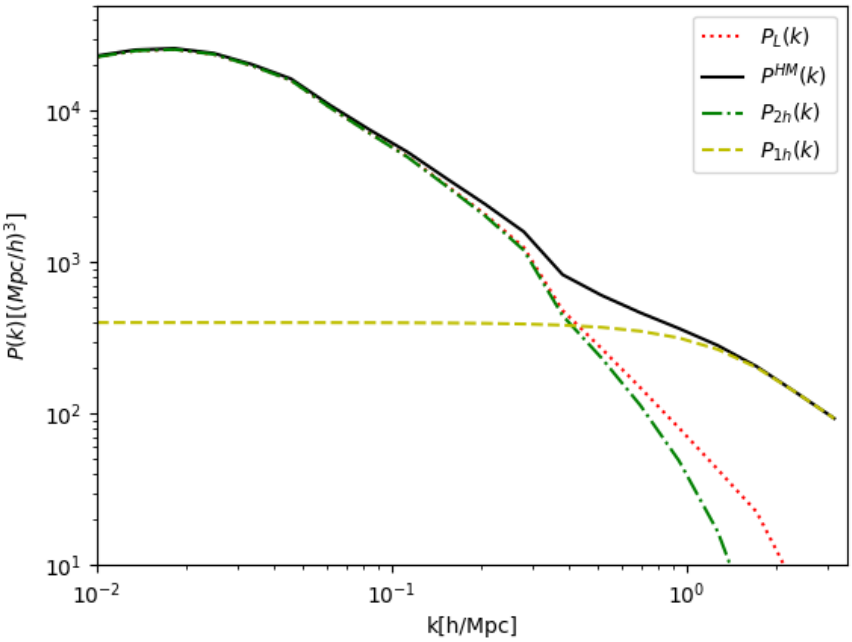


Figure 23: $P^{\mathrm{HM}}(k, a = 1)$

18 星系团

18.1 质量-温度关系

动能：

$$E_k = \frac{1}{2} M \langle v^2 \rangle \tag{169}$$

势能：

$$E_p = -\frac{3}{5} \frac{GM^2}{R_{vir}} \tag{170}$$

处于位力平衡的暗晕：

$$E_k = -\frac{1}{2} E_p \tag{171}$$

热力学平衡下的理想气体：

$$\frac{1}{2} m_p \langle v^2 \rangle = \frac{3}{2} kT \tag{172}$$

暗晕的质量-温度关系：

$$M = 1.38 \times 10^{14} h^{-1} M_\odot \left(\frac{T}{\mathrm{keV}} \right)^{3/2} \left(\frac{\Delta_{\mathrm{vir}}}{200} \right)^{-1/2} \tag{173}$$

通常取暗晕的过密度 $\triangle_{vir} = 200$

18.2 顶帽窗函数下的功率谱

$z=0$ 时，采用顶帽窗函数、质量为 M 且密度扰动超过 δ_{cr} 的星系团/暗晕的功率谱：

$$\begin{aligned} P_{\mathrm{g,thr}}(k,M) &= (b_1^{\mathrm{thr}})^2 P_{\mathrm{L}}(k) W_R^2(k) \\ &+ \frac{1}{2}(b_2^{\mathrm{thr}})^2 \int \frac{d^3p}{(2\pi)^3} P_{\mathrm{L}}(p) W_R^2(p) P_{\mathrm{L}}(|\mathbf{k}-\mathbf{p}|) W_R^2(|\mathbf{k}-\mathbf{p}|) \\ &+ \cdots \end{aligned} \tag{174}$$

其中：

$$\begin{aligned} b_1^{\mathrm{thr}} &= \sqrt{\frac{2}{\pi}} \frac{e^{-\nu^2/2}}{\mathrm{erfc}[\nu/\sqrt{2}]\sigma(M)} \overset{\nu\gg 1}{\approx} \frac{\nu}{\sigma(M)} \\ b_2^{\mathrm{thr}} &= \sqrt{\frac{2}{\pi}} \frac{e^{-\nu^2/2}\cdot\nu}{\mathrm{erfc}[\nu/\sqrt{2}]\sigma^2(M)} \overset{\nu\gg 1}{\approx} \frac{\nu^2}{\sigma^2(M)} \\ \nu &= \delta_{cr}/\sigma(M) \\ R &= R_L(M) \end{aligned} \tag{175}$$

19 引力透镜

19.1 偏折角

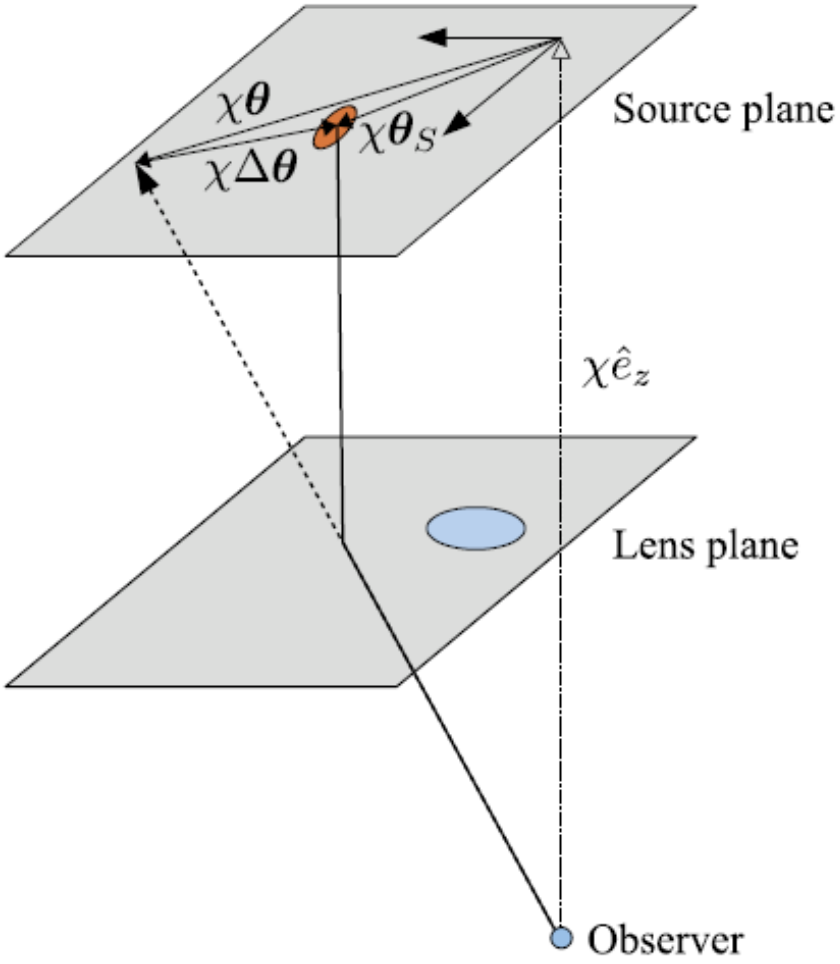


Figure 24: 引力透镜

引力透镜示意如图24
光子测地线方程:

$$\frac{d\hat{p}}{d\chi} = -2(\delta^{ik} - \hat{p}^i\hat{p}^k)\Phi_{,k} \tag{176}$$

透镜势:

$$\phi_L(\theta) = 2\int_0^\chi \frac{d\chi'}{\chi'}\Phi(\theta,\chi')(1-\frac{\chi'}{\chi}) \tag{177}$$

偏折角:

$$\Delta\theta^i = \theta_s^i - \theta^i = \frac{\partial}{\partial\theta^i}\phi_L(\theta) \tag{178}$$

畸变张量:

$$-\Psi_{ij} = -\frac{\partial^2}{\partial\theta^i\partial\theta^j}\phi_L(\theta) = \kappa\delta_{ij} + \begin{pmatrix}\gamma_1 & \gamma_2\\ \gamma_2 & -\gamma_1\end{pmatrix} \tag{179}$$

会聚度：κ；剪切：γ_1, γ_2。
偏折角矩阵：

$$A_{ij} = \frac{\partial\theta_s^i}{\partial\theta^i} = \delta_{ij} + \Psi_{ij} \tag{180}$$

19.2 质点引力透镜

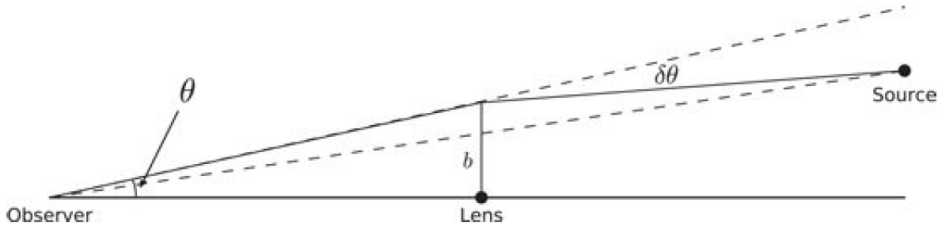


Figure 25: 遥远光源的质点透镜效应

透镜偏折角：

$$\Delta\theta^i = \theta_s^i - \theta^i = \frac{\partial}{\partial\theta^i}\phi_L(\theta) = 2\int_0^\chi \frac{d\chi'}{\chi'}\Phi_{,i}(\theta,\chi')(1-\frac{\chi'}{\chi}) \tag{181}$$

遥远光源的质点透镜效应如图25所示，出自 Dodelson 所著《Gravitational Lensing》。[1]
对于位于 $(0, \chi_{lens})$ 处的质点，由牛顿引力定律可得：

$$\Phi(y = \chi\theta, \chi) = \frac{GM}{\sqrt{y^2 + (\chi - \chi_{lens})^2}} \tag{182}$$

光源极为遥远，$\chi_S \to \infty$，且 $b \ll \chi_{lens}$，

$$\Delta\theta^i = 2\int_0^\infty \frac{d\chi'}{\chi'}\frac{\partial\Phi}{\partial\theta^i}\bigg|_{\chi\theta=b} = 2\int_0^\infty d\chi'\frac{\partial\Phi}{\partial y}\bigg|_{y=b} \tag{183}$$

可得：

$$\Delta\theta = -\frac{4GM}{b} \tag{184}$$

该结果已在爱丁顿日食观测中得到验证。

19.3 星系形状

星系形状张量：

$$q_{ij} = \frac{1}{2}q\delta_{ij} + \frac{1}{2}q\begin{pmatrix}\epsilon_1 & \epsilon_2\\ \epsilon_2 & -\epsilon_1\end{pmatrix} \tag{185}$$

$$q_{ij} = \langle \theta_i \theta_j \rangle_{I_{obs}} = \frac{1}{F} \int d^2\theta I_{obs}(\theta) \theta_i \theta_j \tag{186}$$

总流量：

$$F = \int d^2\theta I_{obs}(\theta) \tag{187}$$

极化张量：

$$I_{ij} = I\delta_{ij} + \begin{pmatrix} Q & U \\ U & -Q \end{pmatrix} \tag{188}$$

由强度守恒：

$$I_{obs}(\theta) = I_{true}(\theta_s) \tag{189}$$

当 $\kappa, \gamma_1, \gamma_2 \ll 1$ 时，可得观测椭率与真实椭率的关系：

$$\begin{aligned} \epsilon_1 &= (1 - 2[\epsilon_1^{true}\gamma_1 + \epsilon_2^{true}\gamma_2])\epsilon_1^{true} + 2\gamma_1 \\ \epsilon_2 &= (1 - 2[\epsilon_1^{true}\gamma_1 + \epsilon_2^{true}\gamma_2])\epsilon_2^{true} + 2\gamma_2 \end{aligned} \tag{190}$$

若真实椭率很小 $\epsilon_1^{true}, \epsilon_2^{true} \ll 1$：

$$\begin{aligned} \epsilon_1 &= \epsilon_1^{true} + 2\gamma_1 \\ \epsilon_2 &= \epsilon_2^{true} + 2\gamma_2 \end{aligned} \tag{191}$$

内禀对齐：

$$q_{ij}^{IA} = -c_1(\eta)\chi^2 \Phi_{,ij}(\mathbf{x}, \eta) \tag{192}$$

19.4 透镜/剪切角功率谱

在傅里叶空间中：

$$\mathbf{l} = l(\cos\theta_l, \sin\theta_l) \tag{193}$$

将剪切 γ_1, γ_2 转换为旋转不变的 E、B 模：

$$\begin{pmatrix} \gamma_1(\mathbf{l}) \\ \gamma_2(\mathbf{l}) \end{pmatrix} = \begin{pmatrix} \cos(2\phi_l) & -\sin(2\phi_l) \\ \sin(2\phi_l) & \cos(2\phi_l) \end{pmatrix} \begin{pmatrix} E(\mathbf{l}) \\ B(\mathbf{l}) \end{pmatrix} \tag{194}$$

在傅里叶空间中：

$$\begin{aligned} E(\mathbf{l}) &= (\frac{l^i l^j}{l^2} - \frac{1}{2}\delta_{ij})(-\Psi_{ij}) \\ B(\mathbf{l}) &= \epsilon_{ik}\frac{l^j l^k}{l^2}(-\Psi_{ij}) \end{aligned} \tag{195}$$

可得：

$$\begin{aligned} E(\mathbf{l}) = \kappa(\mathbf{l}) &= \frac{1}{2}l^2\phi_L(\mathbf{l}) \\ B(\mathbf{l}) &= 0 \end{aligned} \tag{196}$$

星系的透镜/剪切角功率谱见图26

$$
\begin{aligned}
C_{EE}(l) = C_{\kappa\kappa}(l) &= \frac{1}{4} l^4 C_{\phi_L \phi_L}(l) \\
&= (\frac{3}{2}\Omega_{m0} H_0^2)^2 \int_0^\infty d\chi a^{-2}(\chi) g_L^2(\chi) P(k = \frac{l}{\chi}, \eta = \eta_0 - \chi) \\
g_L(\chi) &= \int_\chi^\infty d\chi' (1 - \frac{\chi}{\chi'}) W(\chi') \\
W(\chi) &= \frac{1}{N_g} \frac{dN_g}{d\chi}
\end{aligned}
\tag{197}
$$

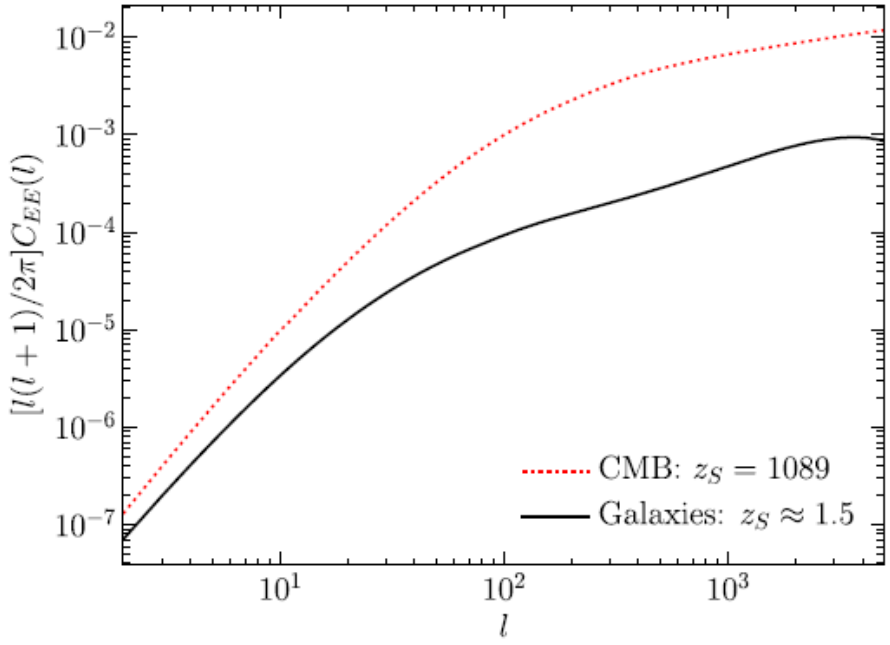


Figure 26: E 模剪切功率谱

将剪切 γ_1, γ_2 转换为旋转不变的切向剪切 γ_t 和交叉剪切 $\gamma_\times$。

剪切关联函数:

$$
\begin{aligned}
\xi_{+,-}(\theta) &= \langle \gamma_t(\mathbf{0}) \gamma_t(\theta) \rangle \pm \langle \gamma_\times(\mathbf{0}) \gamma_\times(\theta) \rangle \\
&= \langle \gamma_1(\mathbf{0}) \gamma_1(\theta) \rangle \pm \langle \gamma_2(\mathbf{0}) \gamma_2(\theta) \rangle \\
&= \int \frac{ldl}{2\pi} J_{0,4}(l\theta) C_{EE}(l)
\end{aligned}
\tag{198}
$$

19.5 剪切交叉角功率谱

投影星系密度:

$$
\Delta_g(\theta) = \int_0^\infty d\chi W_g(\chi) \delta_g(\theta, \eta = \eta_0 - \chi) \tag{199}
$$

在大尺度上，星系-物质交叉功率谱满足:

$$P_{gm}(M,k,z) = b_1(M,z)P(k,z), \quad (k \ll 0.1h/Mpc) \tag{200}$$

剪切交叉角功率谱:

$$C_{gE}(l) = (\frac{3}{2}\Omega_{m0}H_0^2)\int_0^\infty \frac{d\chi}{\chi}W_g(\chi)a^{-1}(\chi)g_L(\chi)P_{gm}(k=\frac{l}{\chi},\eta=\eta_0-\chi) \tag{201}$$

剪切交叉关联函数

$$\begin{aligned}\xi_{g,t}(\theta) = \langle\Delta_g(\mathbf{0})\gamma_t(\theta)\rangle = -\int\frac{ldl}{2\pi}J_2(l\theta)C_{gE}(l)\\ \xi_{g,\times} = \langle\Delta_g(\mathbf{0})\gamma_\times(\theta)\rangle = 0\end{aligned} \tag{202}$$

19.6 CMB 透镜

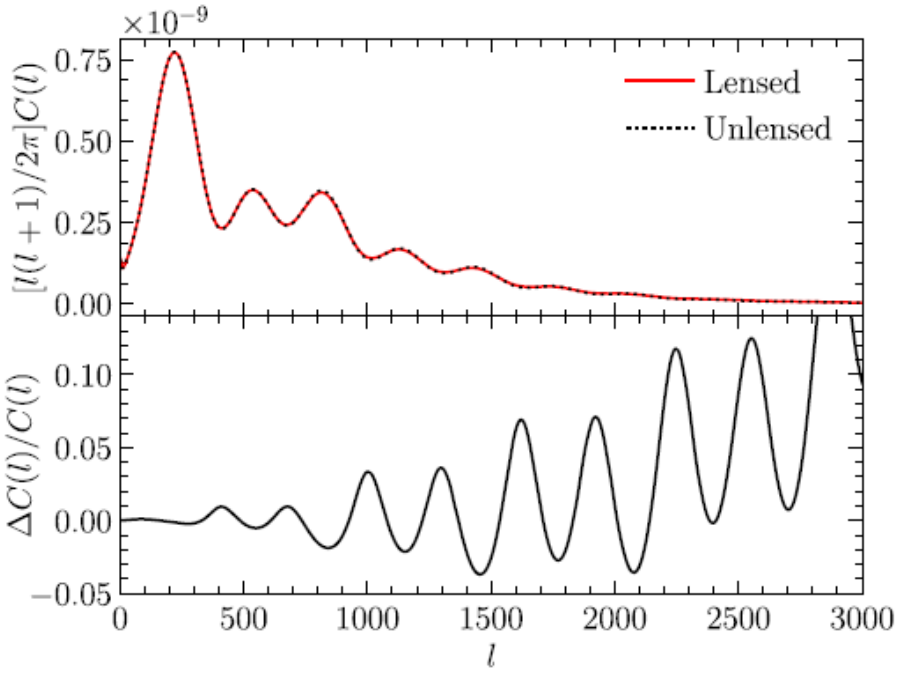


Figure 27: 透镜效应对 CMB 温度功率谱的影响

由引力透镜导致的偏折角可得:

$$T_{obs}(\theta) = T(\theta_s) \tag{203}$$

CMB 的透镜角功率谱:

$$C_{\phi_L\phi_L}(l) = \frac{4}{l^4}(\frac{3}{2}\Omega_{m0}H_0^2)^2\int_0^{\chi_*} d\chi a^{-2}(\chi)(1-\frac{\chi}{\chi_*})^2P(k=\frac{l}{\chi},\eta=\eta_0-\chi) \tag{204}$$

CMB 透镜的角功率谱见图27。

$$
\begin{aligned}
C^{obs}(l) &= (1 - l^2/l_{lens}^2)C(l) + C^{(22)}(l) \\
l_{lens}^{-2} &= \frac{1}{4\pi}\int_0^\infty dl_1 l_1^3 C_{\phi_L\phi_L}(l_1) \\
C^{(22)}(l) &= \frac{1}{(2\pi)^2}\int_0^{2\pi} d\theta \int_0^\infty dl_1 l_1 (l l_1 \cos\theta - l_1^2)^2 C_{\phi_L\phi_L}(l_1) C(\sqrt{l^2 + l_1^2 - 2 l l_1 \cos\theta})
\end{aligned}
\tag{205}
$$

20 数据分析

20.1 概述：从原始数据到参数约束

流程如图28所示

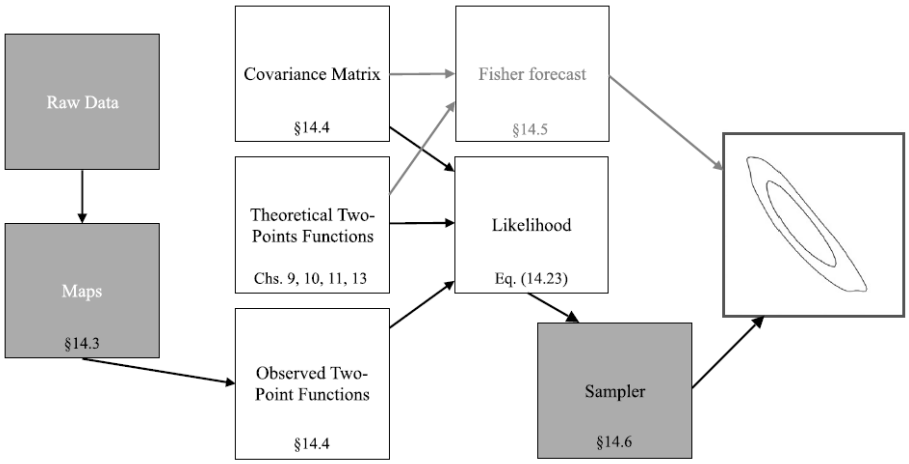


Figure 28: 概述：从原始数据到参数约束

20.2 似然函数

贝叶斯定理：

$$P(w,\sigma_w|\{d_i\}) = \frac{P(\{d_i\}|w,\sigma_w)P(w,\sigma_w)}{P(\{d_i\})} \tag{206}$$

$$P(w,\sigma_w|\{d_i\}) \propto \mathcal{L}\left(\{d_i\}_{i=1}^m \middle| w,\sigma_w\right) P_{\rm prior}(w,\sigma_w) \tag{207}$$

边缘化后验：

$$P(w|\{d_i\}) = \int_0^\infty d\sigma_w P(w,\sigma_w|\{d_i\}) \tag{208}$$

高斯分布的似然函数：

$$\mathcal{L}\left(\{d_i\}_{i=1}^m \middle| w,\sigma_w\right) = P\left(\{d_i\}_{i=1}^m \middle| w,\sigma_w\right) = \frac{1}{(2\pi\sigma_w^2)^{m/2}} \exp\left\{-\frac{\sum_{i=1}^m (d_i-w)^2}{2\sigma_w^2}\right\} \tag{209}$$

为最大化似然函数，令：

$$\frac{\partial\mathcal{L}}{\partial w} = 0, \quad \frac{\partial\mathcal{L}}{\partial\sigma_w^2} = 0 \tag{210}$$

可得：

$$\hat{w} = \frac{1}{m}\sum_{i=1}^m d_i, \quad \widehat{\sigma_w^2} = \frac{1}{m}\sum_{i=1}^m (d_i-w)^2 \tag{211}$$

展开似然函数：

$$\begin{aligned}\ln\mathcal{L}(w) &= \ln\mathcal{L}(\hat{w}) + \left.\frac{\partial\ln\mathcal{L}}{\partial w}\right|_{w=\hat{w}}(w-\hat{w}) + \frac{1}{2}\left.\frac{\partial^2\ln\mathcal{L}}{\partial w^2}\right|_{w=\hat{w}}(w-\hat{w})^2 \\ &= \ln\mathcal{L}(\hat{w}) - \frac{m}{2\sigma_w^2}(w-\hat{w})^2,\end{aligned} \tag{212}$$

可得：

$$\mathrm{Var}[\hat{w}] = \left\langle(\hat{w}-w)^2\right\rangle = -\Big[\frac{\partial^2\ln\mathcal{L}}{\partial w^2}\Big]^{-1} = \frac{\sigma_w^2}{m} \tag{213}$$

$$\mathrm{Var}(\sigma_w^2) \equiv \left\langle\left(\widehat{\sigma_w^2} - \sigma_w^2\right)^2\right\rangle = \frac{2}{m}\sigma_w^4 \tag{214}$$

20.3 示例：CMB 功率谱

对于 CMB 实验，我们通过观测到的温度各向异性 $\hat{C}(l)$ 来推断参数 $\{\lambda_\alpha\}$

我们观测到的 CMB 温度各向异性：

$$\Delta = \frac{T-T_0}{T_0} \tag{215}$$

考虑波束图案 $B(\hat{\mathbf{n}},\hat{\mathbf{n}}')$ 与噪声 $\eta(\hat{\mathbf{n}})$：

$$\Delta(\hat{\mathbf{n}}) = \int d\Omega'\Theta(\hat{\mathbf{n}}')B(\hat{\mathbf{n}},\hat{\mathbf{n}}') + \eta(\hat{\mathbf{n}}) \tag{216}$$

球谐函数展开下：

$$a_{lm}^{\mathrm{obs}} = \int d\Omega Y_{lm}^*(\hat{\mathbf{n}})\Delta(\hat{\mathbf{n}}) = \sum_{l'm'} B_{lm,l'm'}a_{l'm'} + \eta_{lm} \tag{217}$$

若波束各向同性 $B(\hat{\mathbf{n}},\hat{\mathbf{n}'}) = B(\hat{\mathbf{n}}\cdot\hat{\mathbf{n}}')$：

$$a_{lm}^{obs} = a_{lm}B_l + \eta_{lm} \tag{218}$$

假设噪声均值为零，其功率谱为：

$$\langle\eta_{lm}\eta_{l'm'}^*\rangle = N(l)\delta_{ll'}\delta_{mm'} \tag{219}$$

由：

$$a_{lm}|C(l) \sim N(\mu=0,\sigma^2=C(l)) \tag{220}$$

可知：

$$a_{lm}^{obs}|C(l) \sim N(\mu=0,\sigma^2=C(l)B_l^2+N(l)) \tag{221}$$

似然函数：

$$\mathcal{L} \equiv P\left(\{a_{lm}^{\mathrm{obs}}\}|C(l)\right) = \left(2\pi\left[C(l)B_l^2+N(l)\right]\right)^{-(2l+1)/2}\exp\left\{-\frac{1}{2}\sum_{m=-l}^{l}\frac{\left|a_{lm}^{\mathrm{obs}}\right|^2}{C(l)B_l^2+N(l)}\right\}. \tag{222}$$

为最大化 $\mathcal{L}$，令 $d\ln\mathcal{L}/dC(l)=0$，可得：

$$\hat{C}(l)=B_l^{-2}\left(\frac{1}{2l+1}\sum_{m=-l}^{l}\left|a_{lm}^{\rm obs}\right|^2-N(l)\right) \tag{223}$$

$$\mathrm{Var}\left[\hat{C}(l)\right]=\left\langle\hat{C}(l)^2\right\rangle-C(l)^2=\frac{2}{2l+1}[C(l)+N(l)B_l^{-2}]^2 \tag{224}$$

不同 l 模式相互独立：

$$\mathrm{Cov}_{ll'}=\mathrm{Var}\left[\hat{C}(l)\right]\delta_{ll'}=\frac{2}{2l+1}[C(l)+N(l)B_l^{-2}]^2\delta_{ll'} \tag{225}$$

对于高斯波束图案：

$$B_l=e^{-l^2\theta_{beam}^2/2} \tag{226}$$

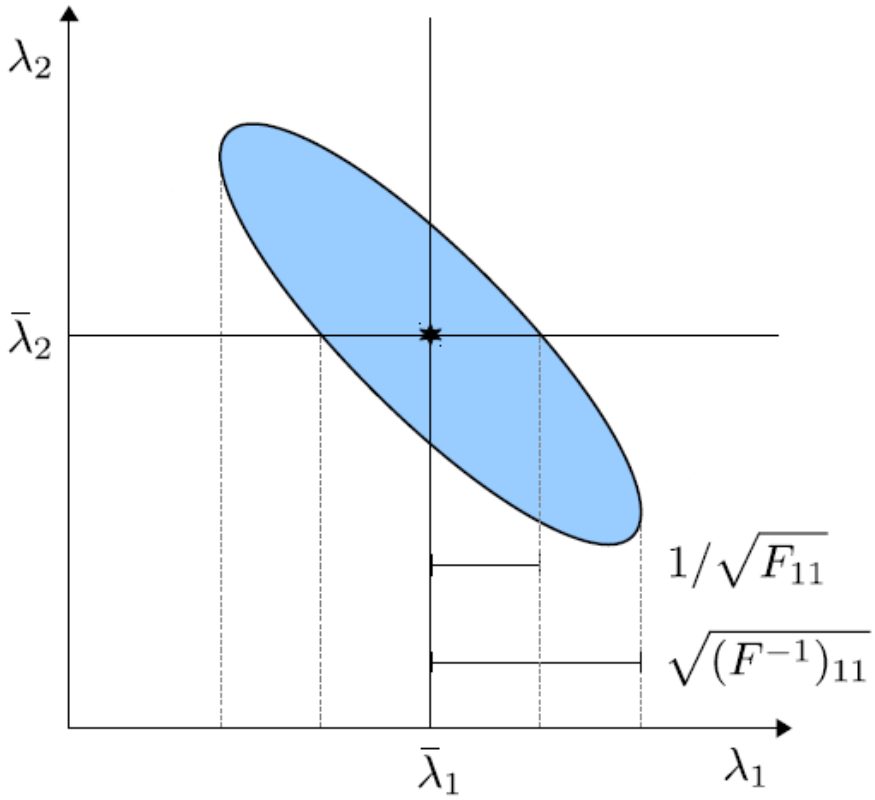


Figure 29: 费雪矩阵给出的误差椭圆

CMB 各向异性的似然函数：

$$\ln\mathcal{L}(\{\lambda_\gamma\})=-\frac{1}{2}\sum_{ll'}\left(\hat{C}(l)-C^{\rm theory}(l,\{\lambda_\gamma\})\right)\left(\mathrm{Cov}^{-1}\right)_{ll'}\left(\hat{C}(l')-C^{\rm theory}(l',\{\lambda_\gamma\})\right) \tag{227}$$

因此，

$$\ln \mathcal{L}(\{\lambda_\gamma\}) = -\frac{\chi^2(\{\lambda_\gamma\})}{2} \tag{228}$$

$$\chi^2(\{\lambda_\gamma\}) = \sum_l \frac{[\hat{C}(l) - C^{\text{theory}}(l, \{\lambda_\gamma\})]^2}{\text{Var}\left[\hat{C}(l)\right]} \tag{229}$$

为最大化 χ^2，令 $d\chi^2/d\lambda_\gamma = 0$，将 χ^2 展开到二阶：

$$\chi^2(\{\lambda_\gamma\}) = \chi^2(\{\bar{\lambda}_\gamma\}) + \sum_{\alpha\beta} F_{\alpha\beta}(\lambda_\alpha - \bar{\lambda}_\alpha)(\lambda_\beta - \bar{\lambda}_\beta) \tag{230}$$

费雪矩阵定义为：

$$\begin{aligned} F_{\alpha\beta} &= \frac{1}{2}\left\langle \frac{\partial^2 \chi^2}{\partial\lambda_\alpha \partial\lambda_\beta} \right\rangle \Bigg|_{\{\lambda_\gamma\}=\{\bar{\lambda}_\gamma\}} \\ &\equiv -\left\langle \frac{\partial^2 \ln \mathcal{L}}{\partial\lambda_\alpha \partial\lambda_\beta} \right\rangle \Bigg|_{\{\lambda_\gamma\}=\{\bar{\lambda}_\gamma\}} \end{aligned} \tag{231}$$

$$(F^{-1})_{\alpha\beta} = \langle (\lambda_\alpha - \bar{\lambda}_\alpha)(\lambda_\beta - \bar{\lambda}_\beta) \rangle \tag{232}$$

对于 CMB 各向异性：

$$F_{\alpha\beta} = \sum_l \frac{1}{\text{Var}\left[\hat{C}(l)\right]} \frac{\partial C^{\text{theory}}(l, \{\bar{\lambda}_\gamma\})}{\partial \bar{\lambda}_\alpha} \frac{\partial C^{\text{theory}}(l, \{\bar{\lambda}_\gamma\})}{\partial \bar{\lambda}_\beta} \tag{233}$$

由费雪矩阵可以绘制参数的误差椭圆，如图29所示。
λ_1 与 λ_2 的联合后验：

$$P(\lambda_1, \lambda_2) \propto \exp\left\{-\frac{1}{2}\lambda_\alpha F_{\alpha\beta}\lambda_\beta\right\} \tag{234}$$

因此，若 λ_2 已知，λ_1 的误差为：$\sqrt{\mathbf{Var}(\lambda_1)} = 1/\sqrt{F_{11}}$
对 λ_2 边缘化后，λ_1 的后验：

$$\begin{aligned} P(\lambda_1) &= \int d\lambda_2 P(\lambda_1, \lambda_2) \\ &\propto \exp\left\{-\frac{\lambda_1^2}{2}\left(\frac{F_{11}F_{22} - F_{12}F_{21}}{F_{22}}\right)\right\} = \exp\left\{-\frac{\lambda_1^2}{2(F^{-1})_{11}}\right\} \end{aligned} \tag{235}$$

因此，若 λ_2 未知，λ_1 的误差为：$\sqrt{\mathbf{Var}(\lambda_1)} = \sqrt{(F^{-1})_{11}}$

20.4 示例：星系功率谱

1. 星系协方差矩阵
   第 i 个像素中的星系数密度扰动：

$$\begin{aligned} \Delta_i &= \int d^3x \psi_i(\mathbf{x})[\frac{n(\mathbf{x}) - \bar{n}(\mathbf{x})}{\bar{n}(\mathbf{x})}] \\ \psi_i(\mathbf{x}) &= \frac{1}{V} e^{i\mathbf{k}_i \cdot \mathbf{x}} \end{aligned} \tag{236}$$

协方差矩阵：

$$\langle \Delta_i \Delta_j^* \rangle = (C_S)_{ij} + (C_N)_{ij} \tag{237}$$

泊松分布粒子带来的散粒噪声 $n_{shot} \sim Pois(\bar{n})$：

$$P(n) = \frac{(\bar{n})^n e^{-\bar{n}}}{n!}, \quad n \in \{0, 1, 2, \dots\} \tag{238}$$

协方差矩阵：

$$\begin{gathered}(C_N)_{ij} = \langle \Delta_i \Delta_j^* \rangle \Big|_{PoissonNoise} = \delta_{ij} \frac{1}{N} \\ N = nV\end{gathered} \tag{239}$$

$$\begin{gathered}(C_S)_{ij} = \langle \Delta_i \Delta_j^* \rangle \Big|_{NoNoise} = \int_0^\infty \frac{dk}{k} [\frac{k^3 P(k)}{2\pi^2}] W_{ij}(k) \\ W_{ij}(k) = \delta_{ij} \frac{9}{2kk_iR^2} \int_{|k-k_i|}^{(k+k_i)R} \frac{dy}{y} j_1^2(y)\end{gathered} \tag{240}$$

当 $k_i R \gg 1$ 时：

$$(C_S)_{ii} = \frac{1}{V} P(k_i) \tag{241}$$

$$\frac{(C_S)_{ii}}{(C_N)_{ii}} = P(k_i)\bar{n} \tag{242}$$

2. 星系功率谱

实空间中，体积 $V = L^3$。

傅里叶空间中，设轴向网格数为 K_{grid}，网格间隔为 $k_F = 2\pi/L$，半轴向总长度为 $k_{N_y} = K_{grid} k_F / 2$。

扰动的傅里叶变换：

$$\begin{gathered}\delta_g(\mathbf{k}) = L^{3/2} \sum_i^{K_{grid}^3} \delta_g(\mathbf{x}_i) e^{-i\mathbf{k}\cdot\mathbf{x}_i} \\ \delta_g(\mathbf{x}) = \frac{1}{K_{grid}^3 L^{3/2}} \sum_{\mathbf{k}}^{k_{N_y}} \delta_g(\mathbf{k}_i) e^{i\mathbf{k}_i\cdot\mathbf{x}}\end{gathered} \tag{243}$$

每个 α 区间包含的模式数，针对一半傅里叶模式（$k_\alpha - \Delta k/2 \le |\mathbf{k}| \le k_\alpha + \Delta k/2$）：

$$m_{k,\alpha} = \frac{1}{2} \frac{4\pi k_\alpha^2 \Delta k}{k_F^3} = \frac{1}{4\pi^2} V k_\alpha^2 \Delta k \tag{244}$$

忽略波束对星系的影响 $B_l = 1$，我们对星系功率谱的估计为：

$$\hat{P}_g(k_\alpha) = \frac{1}{m_{k,\alpha}} \sum_{\mathbf{k}}^{||\mathbf{k}|-k_\alpha|<\Delta k/2} |\delta_g(\mathbf{k})|^2 - P_N \tag{245}$$

δ_g 服从高斯分布，可得标准差：

$$\sqrt{\mathbf{Var}[\hat{P}_g(k_\alpha)]} = \sqrt{\langle \hat{P}_g(k_\alpha)^2 \rangle - \langle \hat{P}_g(k_\alpha) \rangle^2} = \sqrt{\frac{2}{m_{k,\alpha}}} [P_g(k_\alpha) + P_N] \tag{246}$$

协方差矩阵：

$$\mathbf{Cov}_{\alpha\beta} = \sqrt{\langle \hat{P}_g(k_\alpha) \hat{P}_g(k_\beta) \rangle - \langle \hat{P}_g(k_\alpha) \rangle \langle \hat{P}_g(k_\beta) \rangle} = \frac{2}{m_{k,\alpha}} [P_g(k_\alpha) + P_N]^2 \delta_{\alpha\beta} \tag{247}$$

对于离散傅里叶空间场：

$$\langle \delta_g(\mathbf{k}) \delta_g(\mathbf{k}') \rangle = \delta_{\mathbf{k},-\mathbf{k}'} [P_g(k) + P_N] \tag{248}$$

$$\begin{aligned} \delta_g^{obs}(\mathbf{x}_i) &= \tilde{W}(\mathbf{x}_i) \delta_g(\mathbf{x}_i) \\ \delta_g^{obs}(\mathbf{k}_i) &= \sum_{\mathbf{k}_j} \tilde{W}(\mathbf{k}_j) \delta_g(\mathbf{k}_i - \mathbf{k}_j) \end{aligned} \tag{249}$$

$$\tilde{W}(\mathbf{x}_i) = \begin{cases} 1, & \mathbf{x}_i \text{ 位于观测天区} \\ 0, & \text{其他区域} \end{cases} \tag{250}$$

A 数学基础

A.1 特殊函数

勒让德多项式：

$$\mathcal{P}_l(\hat{n}\cdot\hat{n}') = \frac{4\pi}{2l+1}\sum_{m=-l}^{l} Y_{lm}(\hat{n})Y_{lm}^*(\hat{n}') \tag{251}$$

$$\int_{-1}^{1}\frac{d\mu}{2}\mathcal{P}_l(\mu)\mathcal{P}_{l'}(\mu) = \frac{1}{2l+1}\delta_{ll'} \tag{252}$$

球谐函数：

$$\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]Y_{lm}(\theta,\phi) = -l(l+1)Y_{lm}(\theta,\phi) \tag{253}$$

$$\int d\Omega Y_{lm}(\hat{n})Y_{l'm'}^*(\hat{n}) = \delta_{ll'}\delta_{mm'} \tag{254}$$

球贝塞尔函数：

$$\frac{d^2j_l}{dx^2}+\frac{2}{x}\frac{dj_l}{dx}+\left[1-\frac{l(l+1)}{x^2}\right]j_l = 0 \tag{255}$$

$$\int_0^\infty drr^2 j_l(kr)j_r(k'r) = \frac{\pi}{2k^2}\delta_D(k-k') \tag{256}$$

$$e^{i\mathbf{k}\cdot\mathbf{x}} = \sum_{l=0}^{\infty} i^l(2l+1)j_l(kx)\mathcal{P}_l(\mu=\hat{k}\cdot\hat{x}) \tag{257}$$

$$\frac{1}{2}\int_{-1}^{1}d\mu\mathcal{P}_l(\mu)e^{iz\mu} = \frac{j_l(z)}{(-i)^l} \tag{258}$$

贝塞尔函数：

$$J_n(x) = \frac{i^{-n}}{\pi}\int_0^\pi d\theta e^{ix\cos\theta}\cos(n\theta) \tag{259}$$

余弦积分与正弦积分：

$$\begin{aligned}\mathrm{Ci}(x) &= -\int_x^\infty\frac{\cos z}{z}dz,\\ \mathrm{Si}(x) &= \int_0^x\frac{\sin z}{z}dz.\end{aligned} \tag{260}$$

A.2 变换

傅里叶变换：

$$\begin{cases}\Theta(\mathbf{x}) = \int\dfrac{d^3\mathbf{k}}{(2\pi)^3}\Theta(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{x}}\\ \Theta(\mathbf{k}) = \int d^3\mathbf{x}\Theta(\mathbf{x})e^{-i\mathbf{k}\cdot\mathbf{x}}\end{cases} \tag{261}$$

勒让德展开：

$$\begin{cases} \Theta(k,\hat{k},\hat{p}) = \sum_{l}(-i)^l(2l+1)\mathcal{P}_l(\mu=\hat{k}\cdot\hat{p})\Theta_l(k) \\ \Theta_l(k) = \dfrac{1}{(-i)^l}\displaystyle\int_{-1}^{1}\frac{d\mu}{2}\mathcal{P}_l(\mu=\hat{k}\cdot\hat{p})\Theta(k,\hat{k},\hat{p}) \end{cases} \tag{262}$$

实空间球谐展开：

$$\begin{cases} \Theta(\mathbf{x},\hat{p}) = \displaystyle\sum_{l=1}^{\infty}\sum_{m=-l}^{l} a_{lm}(\mathbf{x})Y_{lm}(\hat{p}) \\ a_{lm}(\mathbf{x}) = \displaystyle\int d\Omega\Theta(\mathbf{x},\hat{p})Y^*_{lm}(\hat{p}) \end{cases} \tag{263}$$

傅里叶空间球谐展开：

$$\begin{cases} \Theta_l(\mathbf{k}) = \dfrac{1}{(-i)^l(2l+1)\mathcal{P}_l(\mu=\hat{k}\cdot\hat{p})}\displaystyle\sum_{m=-l}^{l} a_{lm}(\mathbf{k})Y_{lm}(\hat{p}) \\ a_{lm}(\mathbf{k}) = \displaystyle\int d^3\mathbf{x}a_{lm}(\mathbf{x})e^{-i\mathbf{k}\cdot\mathbf{x}} \end{cases} \tag{264}$$

References

[1] Scott Dodelson. *Gravitational lensing.* Cambridge University Press, 2017.

[2] Scott Dodelson and Fabian Schmidt. *Modern Cosmology.* 2020.